Gluing Local Contexts into Global Meaning: A Sheaf-Theoretic Decomposition of Transformer Representations

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Abstract

Prior research into interpretability does not test whether a representation stays consistent under paraphrasing. Each pair of paraphrases can be thought of as a local witness that two activations should be the same. From these local witnesses we can build a cellular sheaf on the paraphrase graph. The question of what is consistent under paraphrasing can be answered by the Laplacian $L_{\mathcal{F}}$ given a frozen model. No training required. Our kernel H^0 gives us a content subspace. We show that with only 20 dims the kernel outperforms the subspace that LEACE retains with all dims on held-out CounterFact fact retrieval by $+17.1$ pp on average across 8 models. By constraining all paraphrases of a fact together we see an additional increase of $+6.5$ pp. Our spectral complement shows us where the causal influence is concentrated. Ablating the coordinates that vary the most when paraphrased (which approximates ablating the spectral complement at the coordinate level) leads to 5.6 to $26.5\times$ larger changes in model outputs compared to ablating coordinates with matched per-coordinate variability. We find this across 9 models from 124M to 13B params. By stacking layers we can create a grid which has $b_1 = N(L-1)$ algebraic 1-cycles. The harmonic mass derived from the model varies by nearly 3 orders of magnitude between the 6 architectures. It also tracks the order of steering fragility (Spearman's $\rho = 0.90$, exact one-sided permutation $p = 0.014$). The kernel, its spectral complement and the harmonic mass all come from the same paraphrase graph. Project page: https://cwru-aism.github.io/gluing-lc-page/.

1 Introduction

Transformers entangle what a sentence means with how it is written. "The Eiffel Tower is in Paris" and "Paris is home to the Eiffel Tower" have different representations even though they mean the same thing. There are techniques that decompose a representation into features [7, 34], see if pre-specified properties are linearly decodable [4], or compare two representation spaces [40, 51]. However none of them deal with consistency between different contexts. That is, if we rephrase a sentence so it keeps the same meaning, will a feature still carry the same information? This becomes an issue when trying to steer models. Some methods may work on certain models and fail or even degrade performance on others [13]. Even within one model, steerability can vary widely with the prompt [56]. There is currently no tool to know what regime your model falls under. We frame this problem as one of local-to-global consistency. Each pair of paraphrases is a local witness that two activations should agree, and together they should form a global consensus.

We create a cellular sheaf over the graph of paraphrases. Here we assign hidden states to the nodes and connect paraphrases with edges. Then we learn a restriction map $P^\top$ ($P \in \mathbb{R}^{d\times k}$) that maps each pair to a common space of dimension k. The coboundary δ^0 gives us the disagreement on each

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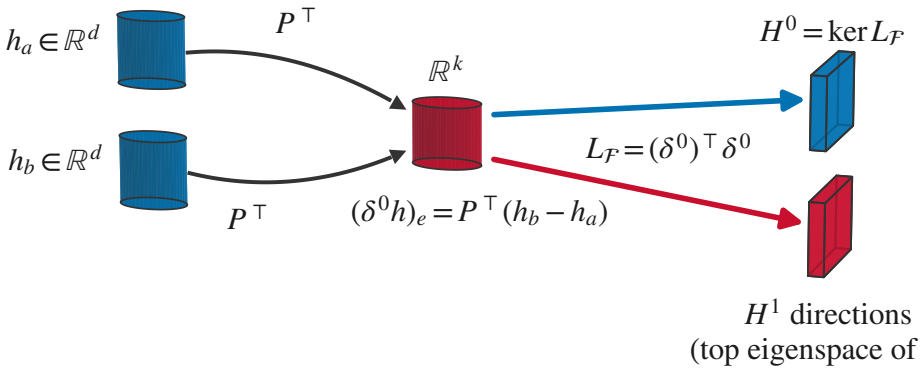


Figure 1: Sheaf decomposition of a paraphrase pair. Hidden states h_a, h_b pass through $P^\top$ into the edge stalk $\mathbb{R}^k$, where the coboundary measures their disagreement, $(\delta^0 h)_e = P^\top(h_b - h_a)$. The bottom and top eigenspaces of the sheaf Laplacian $L_{\mathcal{F}}$ split the edge stalk $\mathbb{R}^k$ into H^0 (blue, content stable across phrasing) and the H^1 directions (red, context-dependent variation).

edge. The sheaf Laplacian $L_{\mathcal{F}} = (\delta^0)^\top \delta^0$ has a kernel $H^0 = \ker(\delta^0)$ of globally consistent sections and a spectral complement of high-disagreement directions. We will refer to these directions as the "H^1 directions" (Remark 3.3 explains the notation).

If we only use one layer, the matching graph has no cycles, so algebraic $H^1 = 0$ and our construction is simply the projected within-pair covariance (Proposition 3.1). If we stack the same activations over multiple layers we get cycles. We can build a grid where each node is a (sentence, layer) pair. Horizontal edges connect paraphrases within each layer and vertical edges connect the same sentence between adjacent layers. This gives $b_1 = N(L-1)$ algebraic 1-cycles (b_1, the first Betti number, is the number of independent cycles), one per square, and each square asks if going from layer l to $l{+}1$ commutes with paraphrase equivalence. The model-derived Hodge harmonic mass (the share of a model-derived edge cochain's energy in the harmonic space, supported on cycles but neither a gradient nor a curl) at $L = 2$ ranges over nearly 3 orders of magnitude across 6 architectures (10^{-4} to 0.08; Table 1) against a 0.25 topological reference, and §5.4 tests it as a diagnostic of steering fragility. When we constrain all paraphrases of a fact together, H^0 outperforms pair-only H^0 by $+6.5$ pp on all 8 architectures. A graph with the same number of edges and no cycle confirms that this improvement comes from constraining the paraphrases jointly, and the cycle itself does not change retrieval (§5.2).

Our contributions are:

- **Content from the kernel.** Sheaf H^0 at 20 dimensions beats LEACE's preserved subspace at full hidden dimension on held-out CounterFact retrieval across eight architectures (mean $+17.1$ pp; 85.3% vs. 43.1% on GPT-2). Random or identity restriction maps keep most of the gain, and constraining all of a fact's paraphrases jointly adds $+6.5$ pp (§5.2).
- **Influence from the spectral complement.** Ablating the most paraphrase-variable coordinates, a coordinate-wise proxy for the top eigenspace of $L_{\mathcal{F}}$, changes model outputs 5.6 to $26.5\times$ more than ablating coordinates of matched per-coordinate variability on nine architectures from 124M to 13B parameters. Projecting out the sheaf H^1 eigenbasis shifts next-token distributions 17.8–$398\times$ more than random subspaces and collapses fact recall to 0–5%, while matched random ablation leaves recall at baseline on the 7B models (§5.3, Appendices B.6 and B.7).
- **Fragility from harmonic mass.** Multi-layer grids carry $b_1 = N(L-1)$ algebraic 1-cycles and a model-derived harmonic mass spanning $[10^{-4}, 0.08]$ that ranks six architectures by steering fragility ($\rho = 0.90$, $p = 0.014$). On Llama-2-7B, contrastive sheaf steering preserves 31.0% of facts against 4.2% for random steering (§5.4).

2 Cellular Sheaves on Paraphrase Graphs

A cellular sheaf $\mathcal{F}$ on a graph $G = (V, E)$ assigns a vector space (stalk) $\mathcal{F}_v = \mathbb{R}^{d_v}$ to each node and a stalk $\mathcal{F}_e = \mathbb{R}^{d_e}$ to each edge, connected by linear restriction maps $\mathcal{F}_{v \trianglelefteq e} : \mathcal{F}_v \to \mathcal{F}_e$ at node-edge

incidences [12, 30]. The coboundary operator $\delta^0 : C^0(\mathcal{F}) \to C^1(\mathcal{F})$ measures local disagreement:

$$(\delta^0 x)_e = \mathcal{F}_{v \trianglelefteq e}(x_v) - \mathcal{F}_{u \trianglelefteq e}(x_u) \tag{1}$$

where $e = (u, v)$ is an edge. If $\delta^0 x = 0$ for all edges then the local data is *globally consistent* and glues into a global section. The sheaf Laplacian $L_{\mathcal{F}} = (\delta^0)^\top \delta^0$ generalizes the graph Laplacian [30]. Figure 1 sketches the pipeline from node stalks through δ^0 to the H^0/H^1 split. Since $x^\top L_{\mathcal{F}} x = \|\delta^0 x\|^2$ we have $\ker(L_{\mathcal{F}}) = \ker(\delta^0) = H^0(\mathcal{F})$ [30], the space of globally consistent sections. The first cohomology $H^1 = \ker(\delta^1)/\mathrm{im}(\delta^0)$ captures obstructions to gluing. For a graph $\delta^1 = 0$, so $H^1 = C^1/\mathrm{im}(\delta^0)$: these are edge functions that are not the coboundary of any node function.

Our interpretation of H^0 is that it contains the information that is preserved through paraphrase. We can think of the spectral complement of $L_{\mathcal{F}}$ as information that varies with context, such as syntax, word choice and surface form [17, 7]. By definition of the kernel, if all paraphrase pairs of a fact agree on a direction, it will stay constant when the fact is paraphrased, and so it tracks the fact's content. The kernel is made up of these directions. Cycles tell us where steering may fail. If the cochain computed from the model does not sum to zero around a cycle (it has a nonzero harmonic component), the model's layer-to-layer maps do not commute with paraphrase equivalence. The fact then travels along inconsistent paths through the network, and the direction we read off one path may change along another. A matching graph (one edge per paraphrase pair) does not have any cycles, so $b_1(G) = 0$, and if P has full column rank k then δ^0 is surjective onto C^1 and we have algebraic $H^1 = \ker(\delta^1)/\mathrm{im}(\delta^0) = 0$. For transformers we place hidden states on the nodes and edges correspond to paraphrase pairs. A restriction map $P^\top$ maps the full activation space to a k-dimensional edge space. The spectral decomposition is taken in edge space, and its eigenvectors are pulled back to the activation space using P. The choice of P (PCA, canonical correlation analysis (CCA), or a contrastive Fisher discriminant) is the only knob, and it changes the decomposition.

LEACE (least-squares concept erasure) [5] learns a concept eraser, which is a good fit for removing the paraphrase / non-paraphrase distinction. For content stability we want a complementary target, so we look for the subspace with the smallest variation within paraphrase pairs. Sparse autoencoders [7, 34] learn features that make up one representation. Centered kernel alignment (CKA) and SVCCA [40, 51] measure how similar two representation spaces are. We want to know if we can split each representation space into a part that is the same across contexts and a part that changes. With this split we can intervene on the directions that change and keep the content. The coboundary gives us this split, and §5 uses these subspaces to intervene.

3 Local Construction on Matching Graphs

Sheaf. The interpretability sheaf $\mathcal{F}$ over a context graph $G = (V, E)$ assigns node stalks $\mathcal{F}(v) = \mathbb{R}^d$, in which the hidden states of one context form a 0-cochain, edge stalks $\mathcal{F}(e) = \mathbb{R}^k$ ($k \ll d$) as a shared comparison space, and linear restriction maps $\mathcal{F}_{v \trianglelefteq e} = P^\top$. The matching graph ($|V| = 2N$, $|E| = N$, no higher simplices) is the simplest case. We learn $P \in \mathbb{R}^{d \times k}$ via three concrete choices: joint PCA (top k components of paired activations), CCA (solving $C_{12} C_{22}^{-1} C_{21} w = \lambda C_{11} w$ for the top k eigenvectors), or contrastive Fisher. The contrastive Fisher map solves $C_W w = \lambda C_T w$ for $C_W = \frac{1}{N} \sum_i \Delta_i \Delta_i^\top$ and $C_T = \frac{1}{N} \sum_i h_i h_i^\top$, where $\Delta_i = h_i^{(1)} - h_i^{(2)}$ is the activation difference of pair i, and takes the k *largest* generalized eigenvectors as columns of P (sign opposite to LDA, orienting P toward maximal within-pair variance relative to total). $L_{\mathcal{F}} = P^\top C_W P$ then has maximal spread, the spectral H^1 complement is more selectively concentrated, and H^0 picks up residual content within the highly variable basis.

Coboundary, Laplacian, decomposition. Orienting each edge as $e = (v_2, v_1)$, with $h^{(1)}$ at v_1, the coboundary $(\delta^0 s)_e = P^\top s_{v_1} - P^\top s_{v_2}$ measures pair disagreement, and the consistency energy $E_{\mathrm{cons}}(s) = \sum_e \|(\delta^0 s)_e\|^2$ vanishes on $H^0 = \ker(\delta^0)$. The empirical sheaf Laplacian in edge space is

$$L_{\mathcal{F}} = \frac{1}{N} \sum_{i=1}^{N} \Delta_i^{(k)} (\Delta_i^{(k)})^\top \in \mathbb{R}^{k \times k}, \quad \Delta_i^{(k)} = P^\top h_i^{(1)} - P^\top h_i^{(2)}. \tag{2}$$

This $k \times k$ matrix equals $\frac{1}{N} Y L_G Y^\top$ where $Y = [P^\top h_v]_v$ and L_G is the Laplacian of the matching graph. It is the second-moment matrix of the edge values of $\delta^0 h$, not the $|V|d \times |V|d$ operator

$(\delta^0)^\top \delta^0$, and we denote it by $L_{\mathcal{F}}$. Given a readout $w \in \mathbb{R}^k$, $w^\top L_{\mathcal{F}} w$ is the Laplacian quadratic form of the observed data seen through $w^\top P^\top$ [29, 20], and minimizing it is a locality preserving projection on the paraphrase graph [31]. Its kernel is the set of readouts under which the data is a global section, i.e. $v \mapsto w^\top P^\top h_v$ is constant along all edges, or equivalently $w^\top \Delta_i^{(k)} = 0$ for all i. We define H^0 to be the span of the lowest m eigenvectors of $L_{\mathcal{F}}$ (a low-energy approximation to $\ker L_{\mathcal{F}}$) and H^1 directions as the span of the highest m. The pullback to the hidden space is $\tilde{v}_j = P v_j$, and the H^0 projector is given by $\Pi_{H^0} = P V_{H^0} V_{H^0}^\top P^\top$ (P has orthonormal columns for the joint-PCA map, so Π_{H^0} is an orthogonal projector; the CCA and contrastive maps of §5.4 have columns with unit norm but are not orthogonalised). The eigenvalue problem is $k \times k$ ($k = 128$) regardless of the hidden dimension, so this step costs the same for 124M models as it does for 13B.

Steering. Given a style direction $\Delta = \bar{h}_{\text{target}} - \bar{h}_{\text{source}}$, we form (a) $\Delta_{H^1} = \Delta - \Pi_{H^0}\Delta$ (H^0-removed) or (b) the projection of Δ onto the top m H^1 dimensions weighted by $s_j = \sum_\ell (\tilde{v}_\ell)_j^2 \lambda_\ell$. All steering vectors are norm-matched to $\alpha = 0.3\|\bar{h}\|$. Baselines: random, full style (contrastive activation addition, CAA [52]), mean-of-differences, variance top-k, Fisher LDA.

The matching-graph limitation. The construction so far gives local consistency: δ^0 measures within-pair disagreement and H^0 contains its kernel. The graph itself, however, has no cycles, and $P^\top$ is surjective (P has full column rank), so the cohomology vanishes above degree zero.

Proposition 3.1 (Reduction to projected within-pair covariance)**.** *With $C_W = \frac{1}{N}\sum_i \Delta_i \Delta_i^\top$ and $C_T = \frac{1}{N}\sum_i h_i h_i^\top$, on a matching graph with one P shared by every node and identity transports on edges: (i) $L_{\mathcal{F}} = P^\top C_W P$; (ii) the Fisher baseline is the generalized eigenproblem $L_{\mathcal{F}} v = \lambda (P^\top C_T P) v$; (iii) $\lambda = \frac{1}{N}\sum_i (\Delta_i^\top P v)^2$ for a unit eigenvector v. Proof: Appendix A.5.*

Remark 3.2 (Per-node restriction maps). With per-node maps P_v and a Procrustes transport Q_{uv} (the orthogonal map that best aligns stalk u to stalk v), the coboundary is $(\delta^0 x)_e = P_v^\top x_v - Q_{uv} P_u^\top x_u$, and $L_{\mathcal{F}}$ is in general not $P^\top C_W P$, nor $P^\top M P$ for any M built from hidden-state second moments alone, because the cross-terms $-P_v^\top h_v h_u^\top P_u Q_{uv}^\top$ carry different maps on the two sides.

Remark 3.3 (Terminology: H^1 directions). Matching graphs are acyclic, so algebraic $H^1 = \ker(\delta^1)/\mathrm{im}(\delta^0) = 0$. We call the top eigenspace of $L_{\mathcal{F}}$ the H^1 *directions* as a mnemonic for high disagreement; it is not a cohomology class. On a matching graph every 1-cochain, $\delta^0 h$ included, is a coboundary, so the name refers to where $\delta^0 h$ carries the most energy. Algebraic $H^1 \neq 0$ appears only on the cycle-bearing topologies of §4.

Proposition 3.1 needs a single P shared across nodes with identity transports. It holds on any graph, and we apply it on the matching graph. Within that regime, the restriction-map ablation (Table 11) places the content in the kernel of the within-pair difference covariance whatever P is, since random or identity maps recover most of the H^0 retrieval; empirically the Fisher baseline of part (ii) degrades fact preservation (15.4% vs. 29.8%). Outside it, multi-layer grids ($b_1 = N(L-1)$) carry a model-derived harmonic mass (§4.1), and per-node P_v with Procrustes transports break the reduction (Remark 3.2, Appendix B.12).

4 Multi-Layer Matching Grids and Algebraic H^1

A single-layer matching graph trivializes algebraic H^1 because it has no cycles and $P^\top$ is surjective. Filling paraphrase cliques does not fix this: each filled clique is simply connected, so H^1 vanishes (its H^2 does not). The most direct source of cycles is network depth.

4.1 Multi-layer matching grids

Let the vertex set be $V = \{(s, l) : s \in \text{sentences}, l \in \text{layers}\}$. Horizontal edges connect paraphrase pairs at each layer, and vertical edges connect each sentence at adjacent layers. The resulting graph (a 1-dimensional complex) is a grid: for N pairs and L layers it has $|V| = 2NL$, $|E| = NL + 2N(L-1)$, and no 2-faces. Direct computation gives $b_1 = N(L-1)$ algebraic 1-cycles. Each elementary cycle is a square (sentence s, paraphrase s') at layers l and $l+1$: the cycle asks whether the layer-$l \to l+1$ transformation commutes with paraphrase equivalence. When it does, ω sums to zero around the square, so ω restricted to that cycle is exact and has no component on the harmonic 1-cochain dual to it; when it does not, ω has a nonzero harmonic component along that cycle.

Table 1: Multi-layer matching grid, $b_1 = N(L-1)$ from depth alone. Hodge harmonic mass of model-derived edge cochains varies by nearly three orders of magnitude. Every row shares the same graph ($b_1 = 30$ at $L = 2$), so the variation comes from the model's layer transitions. 30 MRPC pairs with joint-PCA per layer at shared $k = 8$. The last column is steering fragility: CounterFact fact preservation (%) under random steering (§5.4, Appendix A.3).

Architecture	$L = 2$	$L = 4$	$L = 8$	$L = 16$	**Pres. (%)**
Mistral-7B	0.0001	0.0001	0.0004	0.0008	39.4
Qwen-2.5-7B	0.0001	0.018	0.016	0.096	34.3
Llama-3-8B	0.001	0.002	0.003	0.004	36.0
Phi-2	0.033	0.126	0.109	0.067	27.5
GPT-2 (124M)	0.036	0.107	0.083	–	9.1
Llama-2-7B	0.079	0.071	0.047	0.028	4.2

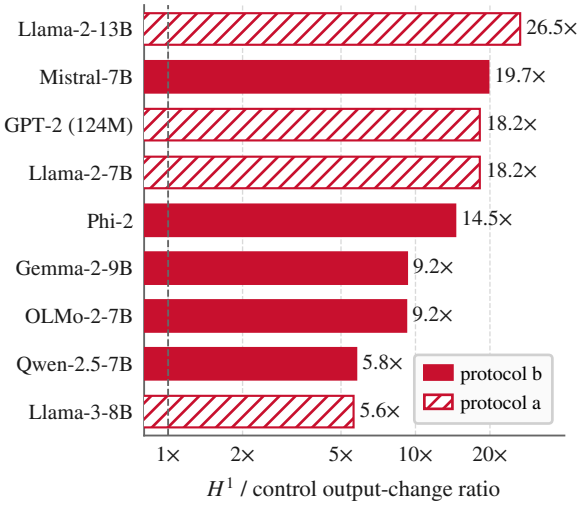


Figure 2: Per-architecture ratio of the output change from ablating the H^1 coordinates to that from ablating control coordinates of matched per-coordinate variability (Table 3). Hatched bars use protocol a (one layer, 64 coordinates, effect on logits) and solid bars protocol b (three depths, 20 coordinates, effect on the final hidden state). Every architecture exceeds the dashed unit baseline, by $5.6\times$ to $26.5\times$.

Definition of the model-derived edge cochain ω. Each layer l has its own joint-PCA restriction map $P_l^\top$ ($P_l \in \mathbb{R}^{d\times k}$) that is common to all sentences. Here h_s^l is the mean-pooled hidden representation of sentence s in layer l. The horizontal edge between paraphrases a, b in layer l carries $\omega(e) = P_l^\top(h_b^l - h_a^l)$. The vertical edge for sentence s from layer l to $l{+}1$ carries $\omega(e) = (P_{l+1}^\top h_s^{l+1} - \mu_{l+1}) - R_{l\to l+1}^\top(P_l^\top h_s^l - \mu_l)$, where μ_l is the mean of the projected activations in layer l and $R_{l\to l+1}$ is the orthogonal Procrustes map between the centred projections of all $2N$ sentences in those layers. If $R = I$ (centring is immaterial), ω would be the coboundary of some node potential and would have no harmonic mass. It's the fitted transport that makes the mass model-dependent. We get $\omega \in \mathbb{R}^{|E|\cdot k}$ by stacking, and the Hodge harmonic mass is $\|\Pi_{\ker L_1}\omega\|^2/\|\omega\|^2$, where $L_1 = \delta^0(\delta^0)^\top + (\delta^1)^\top\delta^1$ is the Hodge 1-Laplacian, which is just the edge Laplacian $\delta^0(\delta^0)^\top$ because the grid has no 2-cells. With all edge stalks identified with one $\mathbb{R}^k$, this is the Hodge decomposition of the constant sheaf $\mathbb{R}^k$ applied to ω coordinate-wise [30, 53]. Because ω is the coboundary of the node potentials $P_l^\top h_s^l - \mu_l$ under the transports $R_{l\to l+1}$, the harmonic component will only come from $I - R_{l\to l+1}^\top$ applied to the within-pair differences. The mass is calculated in the joint-PCA frames P_l. The Hodge decomposition $C^1 = \operatorname{im}\delta^0 \oplus \ker L_1 \oplus \operatorname{im}(\delta^1)^\top$ is orthogonal [43]. Sensitivity to k and to other choices of restriction maps is in Appendix A.2 (robust/fragile split preserved at $k \in \{8, 64, 128\}$).

The Betti number b_1 scales linearly in L from zero, so depth is a single knob that controls the cycle structure, and the model-derived harmonic mass spans $[10^{-4}, 0.08]$ at $L = 2$ (Table 1). A random Gaussian cochain has expected harmonic fraction exactly $b_1/|E| = 0.25$ at $L = 2$; that value is

the random-probe reference. The model's ω places near-zero weight (about 10^{-4}) on the harmonic subspace for Mistral-7B and Qwen-2.5-7B and substantial weight for Llama-2-7B. Scaling k from 8 to 128 and triangulating each square preserve the split between the three low-mass and three high-mass architectures (Appendix A.2). Section 5.4 tests whether the mass predicts steering fragility.

Why the 2-simplices stay unfilled. The grid has no 2-cells, so its cohomology vanishes above degree 1 and, with constant $\mathbb{R}$ coefficients, $\dim H^1 = b_1 = N(L-1)$ ($k\, b_1$ with $\mathbb{R}^k$ stalks). Adding two 2-simplices sharing a diagonal (a, l)-$(b, l+1)$ to each square makes the square cycle the boundary of $T_1 - T_2$ (orientations based on vertex order). Direct computation on this 2-complex gives $b_1 = b_2 = 0$ for all L and N tried, so the harmonic mass is always zero. What remains is the coexact fraction of ω (the curl of ω around the filled triangles), which still separates robust from fragile architectures (Appendix A.2).

4.2 Relation hypergraphs

A second construction yields cycles from relation structure rather than depth. Let $\mathcal{D}$ contain n_f facts with n_p paraphrases each, partitioned into n_r relations. For each fact we add a ring on its n_p paraphrase nodes, and for each relation a ring through the first paraphrase of each of its facts, leaving 2-simplices unfilled. With $n_f = 60$, $n_p = 5$, $n_r = 6$ this gives 300 nodes, $300 + 60 = 360$ edges and no faces, with $\dim H^0 = 6$ and $\dim H^1 = 66$ (Euler $\chi = -60 = 6 - 66$). The 66 classes decompose as 60 within-fact paraphrase rings (one per fact) plus 6 relation-level cycles (one per relation). The matching graph on the same nodes has $\dim H^1 = 0$, and filling fact cliques into 2-simplices also gives $\dim H^1 = 0$, so the relation hypergraph isolates non-trivial topology from the same activations.

Heterogeneous restriction maps and holonomy. Per-node P_v with Procrustes transports $Q_{uv} \neq I$ break Proposition 3.1 (Remark 3.2). Cycle holonomy obstructions on the relation hypergraph are 2.5–3.1, below the Haar RMS value $\sqrt{2k} = 4$, and no architecture deviates significantly from a shuffled-paraphrase null (one-sided $p = 0.10$–0.33, Appendix B.12). A node-space operationalization of H^1 did not transfer to retrieval or steering (Appendix B.13). Both cycle-bearing constructions, depth-induced grid and relation-induced hypergraph, lift the framework off the matching-graph $H^1 = 0$ floor.

5 Experiments

5.1 Setup and Datasets

We test 9 transformer models of varying architectures (GPT-2 [50], Llama [58, 25], Mistral [38], Qwen [65], phi [42], Gemma [23], OLMo [63]) and sizes (124M to 13B parameters). Functional validation ablates coordinates over 1000 sentences (4 models; 666 MRPC and 334 PAWS) or 100 MRPC sentences (5 models). Restriction maps (PCA, CCA, contrastive Fisher) are learned from 200 pairs of the Microsoft Research Paraphrase Corpus (MRPC) [15]. We steer using CounterFact [47] ($n = 1000$ per model) and retrieve facts using ROME CounterFact, with 500 facts split 70/30 for held-out evaluation (§5.2) and a 60-fact, 5-paraphrase subset for the topology constructions. We use mean-pooled hidden states from layer $\ell \approx 2L/3$, set the edge dimension to $k = 128$ to capture $\geq 98\%$ of the paired variance, take the lowest and highest 20 eigenvectors as the H^0 and H^1 directions, and norm-match all steering vectors to $\alpha = 0.3\|\bar{h}\|$. We use bootstrap CIs with 1000 resamples, McNemar's test [46] for paired comparisons, and Cohen's d [11] for effect sizes.

5.2 Content from the Kernel: H^0 Beats LEACE, and Joint Paraphrase Constraints Improve It

Paraphrase-invariance alone does not guarantee fact discriminability: an invariant subspace could conflate Paris–France with Berlin–Germany. Hard same-relation retrieval tests this, against baselines that are low-dimensional or designed for invariance. We compare sheaf H^0 (20D) to LEACE's preserved subspace on ROME CounterFact, with a 70/30 train/test split of 500 facts (n_{test}=150). LEACE erases the binary paraphrase/non-paraphrase concept with an oblique rank-$(d-1)$ projection P^*, and we call $\operatorname{im} P^*$ the LEACE-preserved subspace, at full hidden dim, while sheaf H^0 is 20-dimensional and built without binary labels. Easy retrieval is top-1 over all held-out centroids, and hard retrieval is limited to same-relation distractors. Other subspace and metric-learning baselines

Table 2: Held-out CounterFact retrieval (top-1 accuracy, hard same-relation distractors, $k = 20$ dims, 70/30 train/test split, $n_{\text{test}} = 150$ facts). Sheaf H^0 basis and LEACE eraser are fit on training facts only, and evaluation is on held-out facts not seen during fitting. Across eight architectures sheaf H^0 at 20 dimensions beats LEACE-preserved at full hidden dimension, with gaps ranging from $+1.5$pp (OLMo-2) to $+42.2$pp (GPT-2).

Model	**Full**	H^0 **(20D)**	**LEACE-full**	**LEACE-20D**	**PCA-20**	**Rand-20**	Δ
GPT-2	43.1	**85.3**	43.1	28.9	24.6	39.1	+42.2
Mistral-7B	77.7	**87.1**	77.7	36.8	34.5	59.2	+9.4
OLMo-2-7B	81.3	**82.8**	81.3	40.9	35.7	62.3	+1.5
Phi-2	69.1	**87.1**	69.1	38.8	37.2	55.1	+18.0
Qwen-2.5-7B	49.4	**64.1**	49.4	36.8	37.5	43.7	+14.7
Gemma-2-9B	54.4	**77.9**	54.4	32.3	27.5	43.5	+23.5
Llama-2-7B	68.2	**80.4**	68.2	25.7	22.3	50.7	+12.2
Llama-3-8B	72.7	**88.0**	72.7	28.2	24.4	55.9	+15.3

under the identical protocol are in Appendix B.5: H^0 outperforms every label-free alternative and performs worse than LDA and NCA with fact labels on 5/8 models.

Hard same-relation retrieval (Table 2) shows sheaf H^0 with 20 dimensions beats LEACE-preserved at full hidden dim on all 8 models (mean $+17.1$pp). The $+42.2$pp GPT-2 gap matches the smallest model, since smaller residual streams concentrate paraphrase content into lower-dimensional shared subspaces. Even on the 7–9B models, H^0 with 20 dimensions outperforms LEACE with 3584–4096 dimensions by 1.5–23.5pp.

Dimension-controlled LEACE. LEACE for a binary concept removes a single direction (its null space has rank 1, so the eraser P^* has rank $d - 1$) [5], so the dim-controlled 20D baseline (LEACE-20D, Table 2) is the top-20 PCs of the LEACE-preserved activations. Sheaf H^0 beats LEACE-20D by 27–60 pp on every architecture (mean $+48.0$ pp). LEACE-20D itself lands within 6 pp of PCA-20, confirming the algebraic prediction that a rank-1 erasure leaves the top-PC ranking essentially unchanged. At matched dimensionality the gap is therefore attributable to the decomposition.

The kernel selection drives the result; the projection P is interchangeable. Table 11 ablates H^0 on the same activations under three P regimes: PCA-P (current), random orthonormal P (5 seeds), and identity P (bottom-20 eigenvectors of C_W in full d-dim native space). Mean across eight architectures: PCA-P 81.6%, random-P 74.2%, identity-P 77.5% vs. PCA-20-without-kernel 30.5% and Random-20 51.2%. With identity P this is the bottom eigenspace of C_W itself, so the content lives in the kernel of the within-pair difference covariance whatever P is; PCA reduces the eigenproblem from $\mathbb{R}^d$ to $\mathbb{R}^{128}$ at a cost of at most 7.4 pp. Non-learned restriction maps likewise match learned ones on heterophilic-graph benchmarks [33]; closed-form learned restriction maps [14] are complementary.

Joint paraphrase constraints improve H^0. The result above uses a pair-only graph, in which each fact contributes a chain of paraphrase pairs. Replacing it with a ring over all of a fact's paraphrases ($b_1 = N$ instead of 0), with activations and P unchanged, asks for invariance across every paraphrase of a fact jointly. Across eight architectures ring-H^0 beats pair-H^0 by $+3.2$ to $+9.5$ pp (mean $+6.5$ pp, Table 10); a clique matches the ring within 0.1 pp. An edge-count-matched control attributes the gain to the joint constraint: an acyclic path over six paraphrases, with the ring's five edges and $b_1 = 0$, recovers it ($+7.7$ pp over pairs), and the ring exceeds that path by $+0.5$ pp, significant on none of the eight architectures (Appendix B.3). The cycle-closing difference is the sum of the path differences, so it adds no constraint on exact global sections: H^0 is set by which paraphrases are connected, and cycles enter through the harmonic subspace of §4.1.

5.3 Influence from the Spectral Complement: H^1 Carries Concentrated Causal Influence

We ablate the H^1 directions using a coordinate-axis proxy and compare to controls. At layer ℓ we rank the d residual-stream coordinates by mean absolute difference within pairs $\mathbb{E}|h^{(1)}_{i,j} - h^{(2)}_{i,j}|$ on MRPC. This is the diagonal analogue of C_W and thus a proxy for the top eigenspace of $L_{\mathcal{F}}$ with the identity as the restriction map (Table 11 shows this map recovers most of the H^0 result). The top m coordinates are H^1 and the bottom m are H^0. The controls are the m coordinates in neither

Table 3: Functional validation: output change from ablating the H^1 coordinates over that from ablating control coordinates of matched per-coordinate variability. [a] 1000 sentences (666 MRPC, 334 PAWS), 64 coordinates, one layer ($L/2$), effect on final-position logits. [b] 100 sentences, 20 coordinates, three layers ($L/4$, $L/2$, $3L/4$), effect on the mean-pooled final hidden state, best of the three depths shown. CIs are bootstrap 95%. Cohen's d uses the pooled standard deviation of the two effect distributions.

Model	**Ratio**	**Layer**	**95% CI**	**Cohen's d**
GPT-2 (124M)[a]	18.2×	L6	[17.2, 19.3]	2.34
Phi-2[b]	14.48×	L8	[14.0, 15.0]	13.98
Mistral-7B[b]	19.71×	L8	[19.1, 20.3]	18.45
Qwen-2.5-7B[b]	5.76×	L7	[5.38, 6.12]	6.63
OLMo-2-7B[b]	9.17×	L16	[8.85, 9.49]	12.93
Gemma-2-9B[b]	9.24×	L21	[8.94, 9.54]	22.02
Llama-2-7B[a]	18.2×	L16	[17.3, 19.1]	4.44
Llama-3-8B[a]	5.62×	L16	[5.52, 5.71]	9.08
Llama-2-13B[a]	**26.5**×	L20	[26.1, 26.8]	**14.3**

set with individual variability closest to the average variability of the H^1 coordinates. We zero the chosen coordinates in the layer-ℓ output, continue the forward pass, and measure the L_2 change in the model's output: the logit change at the last position for the 1000-sentence runs ($m = 64$, one layer at $L/2$) and the change in the mean-pooled final hidden representation for the 100-sentence runs ($m = 20$, layers $L/4$, $L/2$, $3L/4$). The ratio is the mean change under H^1 ablation over the mean change under control ablation, with a bootstrap CI. The controls are matched coordinate to coordinate and have only 0.13 to 0.53 of the total variability of the H^1 set. So the ratio compares coordinates of matched variability, and part of it reflects the greater total variability that the H^1 set removes [45, 64]. We use "causal" in the activation patching sense [61, 22, 8]. Projecting out the sheaf H^1 eigenbasis itself causes 17.8–398× more change in the next-token distribution than projecting out a random subspace of the same size across all 9 models (median over layers, Appendix B.6). It also affects the generated text: fact recall collapses under H^1 ablation and stays at baseline under matched random ablations. The H^1 subspace is the least dormant subspace we have measured, which argues against the dormant pathway explanation of the subspace patching illusion [45] (Appendix B.7).

The H^1 coordinates have 5.6–26.5× more influence than the control coordinates on all 9 models, with all 95% CIs excluding 1 (Table 3, Figure 2). GPT-2 (124M) and Mistral-7B have near-identical ratios (18.2× and 19.71×) despite being 56× apart in size. At 3 depths on all 9 models (protocol b), the ratio peaks at $L/4$ for 7 models and at $L/2$ for OLMo-2-7B and Gemma-2-9B (Appendix B.9).

5.4 Fragility from Harmonic Mass: Sheaf Steering as an Architecture Diagnostic

Harmonic mass is computed from MRPC layer-transition differences and steering fragility from CounterFact fact preservation under random perturbation, so the two come from disjoint measurements of the same model. Fragility is the fraction of CounterFact facts preserved under a random direction norm-matched to 0.3 times the hidden-state norm at layer $\lfloor 5L/8 \rfloor$ (Appendix A.3). The three lowest-mass architectures are the three most robust and the three highest are the three most fragile: Spearman $\rho = 0.90$, exact one-sided permutation $p = 0.014$ over the 720 rankings. The one rank inversion, Qwen-2.5-7B against Llama-3-8B, is within sampling error (34.3% vs. 36.0%). Table 1 lists both quantities per model.

The rest of this subsection compares steering methods within a model. CounterFact evaluates whether steering preserves facts while changing style. We compute style direction as the difference in mean activations between target and original completions at layer $\ell = \lfloor 5L/8 \rfloor$, then add norm-matched perturbations. Style change is measured as the fraction of outputs differing from baseline, and fact preservation is the fraction containing the target entity (greedy decoding, 50 tokens). At $\alpha = 0.3$, style saturates at 100% for all methods on all models, so fact preservation is the key differentiator.

Table 4 shows an architecture-dependent pattern. Llama-2-7B is fragile, and random steering collapses fact generation (4.2%), while directed methods retain facts at 12–31%, led by the architecture-matched contrastive sheaf at 31.0% (McNemar $p < 10^{-70}$ vs. random). Mistral-7B is robust, and

Table 4: Steering on CounterFact ($\alpha = 0.3$, $n = 1000$ per model). Fact preservation (%) with 95% bootstrap CIs. McNemar p is paired against the best method in each block (bold). Style saturates at 100% for all methods on all models.

Model	Method	Fact%	95% CI	McNemar p
Llama-2-7B	Contrastive sheaf (H^1 dims)	**31.0**	[28.2, 33.9]	–
	PIXEL [66]	21.1	[18.6, 23.9]	$< 10^{-16}$
	Full style (CAA)	13.7	[11.7, 15.9]	$< 10^{-31}$
	H^0-removed (sheaf)	13.7	[11.6, 15.9]	$< 10^{-31}$
	H^1 dims (sheaf PCA)	12.1	[10.0, 14.1]	$< 10^{-38}$
	Random	4.2	[3.0, 5.4]	$< 10^{-70}$
Llama-3-8B	PIXEL [66]	**38.8**	[35.2, 42.4]	–
	Random	36.0	[33.0, 38.8]	0.10
	Full style (CAA)	27.2	[24.5, 29.9]	$< 10^{-15}$
	H^1 dims (sheaf PCA)	26.1	[23.4, 28.7]	$< 10^{-2}$
	H^0-removed (sheaf)	26.1	[23.3, 28.8]	$< 10^{-2}$
Mistral-7B	CCA sheaf (H^1 dims)	**41.6**	[38.6, 44.4]	–
	Random	39.4	[36.3, 42.4]	0.19
	PIXEL [66]	34.8	[31.1, 38.3]	$< 10^{-8}$
	Variance (top-k)	33.0	[30.0, 35.8]	$< 10^{-7}$
	Full style (CAA)	31.7	[28.8, 34.6]	$< 10^{-9}$
	H^1 dims (sheaf PCA)	29.8	[26.9, 32.6]	$< 10^{-12}$
Llama-2-13B	H^1 dims (sheaf PCA)	**32.7**	[29.8, 35.6]	–
	Random	32.1	[29.3, 35.0]	0.74
	Variance (top-k)	31.3	[28.5, 34.1]	0.82
	Full style (CAA)	29.6	[26.8, 32.3]	0.07

the architecture-matched CCA sheaf has the highest value (41.6% vs. 39.4% for random, McNemar $p = 0.19$, so they are statistically tied). Llama-3-8B is also robust, with random and PIXEL having the highest values (36.0–38.8%). Random can compete at higher α on robust models, which is consistent with the concentrated influence of the H^1 coordinates (§5.3) disrupting more at large magnitudes. Llama-2-13B falls in between, all methods performing similarly (29.6–32.7%, McNemar $p > 0.07$). Consistent with this, the sheaf directions outperform random at lower magnitudes (see below).

Operating regime. We present $\alpha = 0.30$ as a stress test. Over a sweep of $\alpha \in [0.05, 0.50]$ (Appendix B.2), H^0-removed sheaf steering with the architecture-matched map outperforms random on both swept models (Mistral-7B and Llama-3-8B) at $\alpha \in [0.10, 0.15]$ (averaging $+1.0$–2.0 pp higher, with H^0-removed reaching 43% vs. 41% for random on Mistral-7B at $\alpha = 0.10$), and random becomes competitive from $\alpha = 0.30$. Use $\alpha \in [0.10, 0.15]$ on robust models and $\alpha \approx 0.30$ on fragile ones.

Head-to-head with PIXEL. PIXEL [66], which extracts steering directions from two views of the prompt and calibrates the magnitude per position, reproduces the architecture-dependent pattern (Table 4): 21.1% on Llama-2-7B (vs. 4.2% random) but 34.8% on Mistral-7B where random reaches 39.4%. PIXEL beats CAA on Llama-2-7B ($p < 10^{-7}$), and contrastive sheaf leads both at 31.0%. The fragility pattern holds for both direction recipes, and our simplifications of PIXEL appear in Appendix B.11.

6 Related Work

This paper extends a non-archival workshop version [24] that contained only the single-layer construction with three pilot experiments; the multi-layer grids, relation hypergraphs ($b_1 = 66$), harmonic-mass fragility diagnostic, held-out LEACE protocol, restriction-map ablation, PIXEL comparison, baseline suite, and self-supervised learning (SSL) connection are new.

Sheaves in machine learning. Hansen and Ghrist [30] developed the spectral theory of the sheaf Laplacian, which specializes to the connection Laplacian of Singer and Wu [55] on $O(d)$-bundles, and

Hansen and Gebhart [28] introduced sheaf neural networks, since applied to heterophilic graphs [6, 2, 33], predictive coding [54], and restriction-map learning [14]. Duta et al. [16] build sheaf hypergraph networks, and Mule et al. [48] unify directed/undirected hypergraph Laplacians. In the closest concurrent work on transformers, Krasnovsky [41] proposes a sheaf-inconsistency score for circuit coherence and Javidnia [37] builds context-chart atlases with Fisher metrics and holonomy. We differ in stalks (paraphrase-graph nodes carrying hidden states under PCA/CCA/contrastive restrictions) and in target (post-hoc H^0/H^1 decomposition for retrieval and steering).

Subspace interpretability. Sparse autoencoders [7, 34, 57, 19], probes [4], causal abstraction [21], CKA/SVCCA [40, 51], and distributed alignment search (DAS) and LEACE [22, 5] target individual features or whole-space comparison. The $\log\det\mathrm{Cov}$ regulariser of §7 shares its geometry with the decorrelation terms of VICReg [3] and Barlow Twins [67]. Invariance objectives (contrastive: SimCLR [9], InfoNCE [60]; non-contrastive: BYOL [26], Barlow Twins [67]; DCCA [1]) are training-time analogues that achieve paired-view agreement during training, while we read invariance off a frozen model via $\ker(L_{\mathcal{F}})$.

Steering and linear representations. Activation addition [59], representation engineering [69] and their variants [66, 62, 39] add directions and are compared in a unified evaluation by Im and Li [35]. Da Silva et al. [13] show across 36 models that steering reliability depends on the model, and Tan et al. [56] show that it's unreliable across inputs, which §5.4 connects to harmonic mass. The linear representation hypothesis [49, 27, 32] is qualified by Engels et al. [18] and Makelov et al. [45]. Our per-coordinate variability-matched controls only partially address this concern (§5.3). We compare to PIXEL (§5.4, Appendix B.11). Angular Steering [62] and MSRS [35] are application-side and orthogonal.

7 Discussion

Proposition 3.1 reduces $L_{\mathcal{F}}$ to projected covariance for a single shared P with identity transports, here on the matching graph. Within that regime the restriction-map ablation locates content in the kernel of the within-pair difference covariance (Table 11). Beyond it, multi-layer grids carry a model-derived harmonic mass whose ranking tracks steering fragility (§5.4). The mass needs only 30 MRPC pairs and one forward pass per sentence, so it can screen a model before a steering sweep. Graphs that join activations of different dimension (text and image encoders, or models of different width) need per-node maps P_v of different widths, which we leave to future work.

Self-supervised connection. A sheaf-regularized loss $\mathcal{L} = \mathrm{tr}(L_{\mathcal{F}}) - \beta \log\det\mathrm{Cov}(f)$ combines VICReg's invariance term ($\mathrm{tr}\, L_{\mathcal{F}}$ is the within-pair MSE up to scale) with a log-det spread term. Like VICReg's variance and covariance terms [3] and Barlow Twins' redundancy reduction [67], this penalizes dimensional collapse, though not in VICReg's hinge-plus-off-diagonal form. Such regularizers are used alongside non-contrastive losses such as BYOL and SimSiam [26, 10]. Across 7 architectures it holds encoder rank at 57–60/64, where MSE alone collapses to 14–42/64 (Appendix B.14).

Limitations. P is architecture-dependent and partially domain-dependent (Appendix B.1). The operating regime is $\alpha \in [0.10, 0.15]$. Node-space H^1 does not transfer, and cycle holonomy is not significantly different from a shuffled null (Appendices B.13, B.12). Our fragility diagnostic is a rank correlation across 6 architectures ($\rho = 0.90$), and we would need more architectures to use it as a predictor. All the construction needs are contexts asserted to be equivalent to each other, and a cross-lingual probe shows it transfers to translation pairs (Appendix B.8). Logical equivalences and dialogue continuations fit the same template, but we have not tried them.

8 Conclusion

By framing cross-context consistency as a sheaf condition, we can view each paraphrase pair as a local witness of agreement. The kernel of the sheaf Laplacian glues these witnesses into a content subspace, and the spectral complement holds the information that changes with phrasing, all read off a frozen model without training. The kernel beats the preserved subspace of LEACE on held-out retrieval, the spectral complement concentrates causal influence, and the cycles the same operator exposes between layers give a harmonic mass whose ranking tracks steering fragility across six architectures ($\rho = 0.90$).

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A Theoretical Foundations

A.1 Algebraic $H^1 \neq 0$ on Richer Topologies

On the matching graph from the main paper, $H^1 = 0$ algebraically because the graph is acyclic and $P^\top$ is surjective. To generate topologies with $H^1 \neq 0$, we assemble three complexes from the same 300 activation vectors (60 facts, 5 paraphrases each):

1. Matching graph (paper baseline): 300 nodes, 120 edges (one per paraphrase pair), 0 faces. Algebraic $H^1 = 0$.
2. Fact clique (filled): 300 nodes, 600 edges (all pairs), 600 filled triangles. Each clique is simply connected, so $H^1 = 0$ (though $H^2 = 240$).
3. Relation hypergraph (unfilled cycles): 300 nodes, 360 edges (a five-node ring per fact, 300 edges, plus a ring per relation through the first paraphrase of each of its ten facts, 60 edges), 0 filled faces. Creates non-contractible cycles, yielding $\dim H^1 = 66$.

Relation rings join facts that share a relation (e.g., all `country-capital` facts). This complex carries cycles that no 2-simplex fills. Cohomology dimensions here use trivial $\mathbb{R}$ coefficients on the underlying simplicial complex; these may differ from the sheaf cohomology of the induced learned-Q sheaf with learned Procrustes transports, and the cycle holonomies in Appendix B.12 are a separate probe of that twist.

Hodge harmonic fraction. On random unit edge cochains (a uniformly random direction for each edge), we compute the fraction of cochain energy in $\ker((\delta^0)^\top)$, the orthogonal complement of $\mathrm{im}(\delta^0)$, and in the harmonic part $\ker((\delta^0)^\top) \cap \ker(\delta^1) \cong H^1$. The two fractions are equal on face-free complexes ($\delta^1 = 0$). For a random cochain we expect the harmonic fraction to be exactly $b_1/|E|$.

The relation hypergraph has expected harmonic fraction exactly $b_1/|E| = 66/360 = 18.33\%$, regardless of architecture; the measured 18.4% is within probe noise. The filled cliques place 60% of

Table 5: Genuine cohomology and Hodge harmonic fraction across architectures and topologies.

Topology	H^0	H^1 (algebraic)	H^2	$\ker(\delta^0)^\top$ **frac.**	**Harmonic frac.**
Matching (acyclic)	180	0	0	0.000	0.000
Clique (filled)	60	0	240	0.600	0.000
Relation hypergraph	6	**66**	0	0.184	**0.184**

their energy outside $\mathrm{im}(\delta^0)$, but all of it is coexact (every cycle is a sum of triangle boundaries), so their harmonic fraction is zero.

Six-model LEACE table (whole 60-fact set).

Table 6: LEACE vs sheaf H^0 on 60-fact retrieval (20 dimensions, top-1 against all 60 facts). LEACE is fitted on paraphrase versus non-paraphrase fact prompts, and sheaf H^0 uses the joint-PCA restriction map.

Model	**Full**	H^0 (20D)	**LEACE preserved**	**LEACE erased**	**PCA-20**	**Random-20**
GPT-2	85.3%	**95.7**%	85.0%	2.0%	22.7%	53.7%
Phi-2	97.3%	93.0%	97.3%	3.3%	38.0%	64.3%
Qwen-2.5-7B	73.3%	**82.0**%	75.0%	1.0%	30.3%	50.0%
OLMo-2-7B	100%	97.0%	100%	2.3%	36.7%	77.3%
Mistral-7B	100%	80.7%	100%	2.0%	50.0%	71.3%
Gemma-2-9B	99.3%	83.7%	99.3%	1.3%	26.7%	68.3%

On the same 300 vectors, Table 6 compares sheaf H^0 with LEACE. Sheaf H^0 outperforms LEACE preserved by up to 10.7pp on GPT-2 and Qwen-2.5-7B and outperforms full-space on those models while only using 20 of 768–3584 dimensions. LEACE makes the smallest least-squares edit that makes a binary concept (paraphrase vs non-paraphrase) linearly unpredictable; we instead identify the directions of minimal within-pair variation. LEACE-erased correctly destroys information (1–3%), which validates the comparison.

A.2 Simplicial Complex Variants and Scale-Up

In the main text we use 1-skeletons and edge dim $k = 8$. We test whether these numbers depend on either choice. Here we fill the multi-layer grid into a 2-complex, fill in the paraphrase cliques of the relation hypergraph, and then recompute Table 1 for $k = 64$ and $k = 128$.

Triangulated multi-layer grid. In Section 4.1 we show that filling in each square with two 2-simplices will cause b_1 and the harmonic mass to be zero. Here we compute these numbers on the resulting 2-complex.

Each diagonal edge from (a, l) to $(b, l{+}1)$ will have $\omega(e) = (P_{l+1}^\top h_b^{l+1} - \mu_{l+1}) - R_{l\to l+1}^\top (P_l^\top h_a^l - \mu_l)$, which is the vertical-edge formula applied across that pair. The architecture-dependent quantity that remains is the *coexact fraction*. This is the energy of the orthogonal projection of ω onto $\mathrm{im}\,(\delta^1)^\top$, which is determined by the curl $\delta^1\omega$. This is the model-implied 2-cochain over the filled-in triangles, or equivalently the curl (circulation) of ω around each triangle we filled in, i.e. the mismatch between the diagonal-then-horizontal and vertical-then-diagonal path sums.

The coexact fraction in Table 7 recovers the partitioning from Table 1 for all values of $L \leq 8$. The three robust architectures are less than 3% while the three fragile ones are between 5% and 20%. So we see the same information that was captured by the harmonic mass of the 1-skeleton is now captured by the coexact direction when we fill the grid into a 2-complex. The layer transitions still do not commute with paraphrase equivalence on the same set of architectures, but this failure is absorbed by the 2-skeleton as a coexact component instead of forming a harmonic class.

Triangulated relation hypergraph. By filling in all $\binom{5}{3} = 10$ triangles for each fact's paraphrase 5-clique we have made each fact simply connected, so the 60 within-fact 1-cycles become 1-boundaries. So we now have $b_0 = 6$ (one component per relation), $b_1 = 6$ (these are the only cycles that go through multiple facts, i.e. the relation cycles) and $b_2 = 240$ (each K_5 2-skeleton has $b_2 = 4$ and

Table 7: Triangulated multi-layer grid (each square filled with two triangles sharing a diagonal). Edge dim $k = 64$, 30 MRPC paraphrase pairs. $b_1 = b_2 = 0$ for every architecture and every L, so ω has no harmonic component (Hodge harmonic mass collapses to machine precision). Reported quantities: model-derived ω norm squared, coexact fraction (energy explained by the 2-skeleton), and face-cochain norm squared $\|\delta^1\omega\|^2$. Mistral-7B and Llama-3-8B place under 1% on the 2-skeleton and Qwen-2.5-7B under 3%, while GPT-2, Phi-2 and Llama-2-7B place 5–20%.

	$L = 2$		$L = 4$		$L = 8$	
Architecture	coexact	$\|\delta^1\omega\|^2$	coexact	$\|\delta^1\omega\|^2$	coexact	$\|\delta^1\omega\|^2$
Mistral-7B	0.0000	1.4×10^{2}	0.0002	8.6×10^{2}	0.0006	2.5×10^{3}
Llama-3-8B	0.0005	4.7×10^{2}	0.0026	1.8×10^{3}	0.0075	5.1×10^{3}
Qwen-2.5-7B	0.0003	9.8×10^{2}	0.0255	6.0×10^{5}	0.0247	9.7×10^{5}
GPT-2 (124M)	0.0544	6.7×10^{3}	0.1219	1.4×10^{5}	0.1014	2.2×10^{5}
Phi-2	0.0572	5.1×10^{3}	0.1895	7.9×10^{4}	0.1970	2.0×10^{5}
Llama-2-7B	0.0759	6.4×10^{4}	0.0815	8.6×10^{4}	0.0623	1.3×10^{5}

there are 60 facts). We find that random unit edge cochains have a Hodge harmonic mass of 0.93% on this 2-complex (expectation of $6/660 = 0.91\%$). This is lower than the 18.4% we found on the 1-skeleton ring (Table 5). By construction the Hodge mass is a function of the topology and $k = 64$ stalk dim and will be the same for all architectures.

There are 6 harmonic 1-forms left, which correspond to the relation cycles. There are $b_2 = 240$ new harmonic 2-forms that do not pull back to anything we can interpret on the residual stream. Under the restriction maps we used with shared PCA, the model-implied 2-cochain $\delta^1\omega$ is 0 to machine precision (all transports are the identity map, so all cochains are exact w.r.t. the node potential $P^\top h_v$). So with these restriction maps the 2-complex carries no degree-2 cohomological obstruction beyond what the 1-skeleton already encodes. If we use per-node restriction maps and learn the Procrustes transports we may see different results. We leave this heterogeneous-restriction H^2 probe to future work.

Edge-dimension scale-up. The body Table 1 uses $k = 8$. We check whether the differences between architectures, their order by Hodge mass, and the body numbers persist at larger k. We repeat the multi-layer grid construction at $k = 64$ ($N = 30$) and $k = 128$ ($N = 80$, as k is bounded by $\min(2N, k_{\text{requested}})$).

Table 8: Multi-layer matching grid Hodge harmonic mass over edge dimension. The body Table 1 shows $k = 8$. As we increase k, the 3 robust architectures stay below the 3 fragile ones for all $L \leq 8$, and the range widens to over 3 orders of magnitude. The $k = 128$ row uses $N = 80$ paraphrase pairs to allow for the requested k (otherwise k is capped at $\min(2N, k_{\text{req}}) = 60$).

	$k = 64$ ($N = 30$)			$k = 128$ ($N = 80$)		
Architecture	$L{=}2$	$L{=}4$	$L{=}8$	$L{=}2$	$L{=}4$	$L{=}8$
Mistral-7B	0.00003	0.0002	0.0005	0.00002	0.0002	0.0005
Llama-3-8B	0.0005	0.0022	0.0061	0.0004	0.0017	0.0049
Qwen-2.5-7B	0.0003	0.0234	0.0248	0.0003	0.0191	0.0215
GPT-2 (124M)	0.050	0.112	0.101	0.041	0.081	0.084
Phi-2	0.054	0.180	0.197	0.053	0.156	0.180
Llama-2-7B	0.067	0.077	0.065	0.083	0.093	0.078

For all $L \leq 8$ and k we see that the robust set of models (Mistral-7B, Llama-3-8B, Qwen-2.5-7B) remains lower than the fragile set of models (GPT-2, Phi-2, Llama-2-7B). Mistral-7B is always the lowest (equal to Qwen-2.5-7B at $k = 8$, $L = 2$). The span widens to over 3 orders of magnitude ($\approx 2 \times 10^{-5}$ to 8×10^{-2}). Within the fragile set, the order changes depending on k and L: Llama-2-7B is the highest at $L = 2$ and Phi-2 is the highest at $L \geq 4$. The body $L = 2$ values are consistent at all edge dimensions ($\approx 10^{-4}$ for Mistral-7B and Qwen-2.5-7B, $\approx 10^{-3}$ for Llama-3-8B and ≈ 0.08 for Llama-2-7B).

A.3 Harmonic Mass vs Steering Fragility (Per-Model)

Table 1 shows the two quantities we use for our test in §5.4. Multi-layer Hodge harmonic mass ($L = 2$) and steering fragility. Here fragility is fact preservation on CounterFact when steered in a random direction with norm equal to 0.3 times the hidden-state norm at layer $\lfloor 5L/8 \rfloor$ (lower is more fragile), with $n = 1000$ prompts (200 for Phi-2), using the same metric as Table 4. Note these quantities are derived from disjoint measurements (MRPC between layers and CounterFact steering). The rank correlation is $\rho = 0.90$ ($p = 0.014$, exact one-sided permutation test over 720 rankings). The only place the ranks disagree is Qwen-2.5-7B and Llama-3-8B, which is within sampling error (34.3% vs. 36.0% with $n = 1000$).

A.4 Geometric Picture: Where H^1 Lives

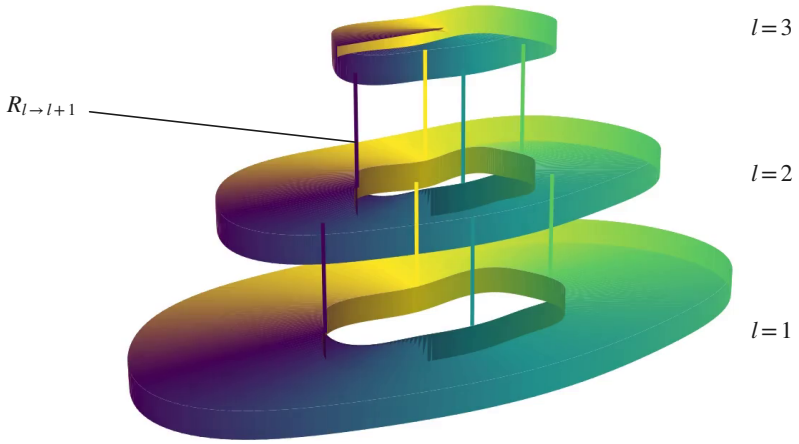


Figure 3: Multi-layer matching sheaf schematic, a frame from an animation. Each slab represents a layer l, with struts connecting the same sentence between adjacent layers via the Procrustes transport $R_{l\to l+1}$. Traveling up one strut, across layer $l+1$ and down another strut closes an inter-layer square. There are $b_1 = N(L-1)$ such independent squares, and they carry the harmonic mass. Slab shapes are arbitrary and do not represent any topology.

A.5 Proof of the Reduction and Spectral Validation

Proof of Proposition 3.1. On the matching graph with a single shared P and identity transports, each edge cochain is $\Delta_i^{(k)} = P^\top h_i^{(1)} - P^\top h_i^{(2)} = P^\top \Delta_i$ with $\Delta_i = h_i^{(1)} - h_i^{(2)}$. Hence

$$L_{\mathcal{F}} = \frac{1}{N}\sum_i \Delta_i^{(k)}(\Delta_i^{(k)})^\top = P^\top\Big(\frac{1}{N}\sum_i \Delta_i\Delta_i^\top\Big)P = P^\top C_W P,$$

which is (i). For unit v, $v^\top L_{\mathcal{F}} v = \frac{1}{N}\sum_i (v^\top P^\top \Delta_i)^2$, which is (iii). The Fisher baseline replaces the standard eigenproblem with the generalized one $L_{\mathcal{F}} v = \lambda(P^\top C_T P)v$, which is (ii). □ For Remark 3.2: with per-node maps the edge cochain is $c_e = P_v^\top h_v - Q_{uv}P_u^\top h_u$, and $\frac{1}{N}\sum_e c_e c_e^\top$ contains the cross-terms $P_v^\top h_v h_u^\top P_u Q_{uv}^\top$, which do not factor through a single P.

Numerical sanity check. Using exact arithmetic, the dimension of the kernel of the assembled Laplacian $(\delta^0)^\top\delta^0$ over C^0 is $\dim\ker = \dim C^0 - \operatorname{rank}(\delta^0)$. We find that the eigenvalue-threshold estimate matches the rank calculation, up to one eigenvalue near the cutoff, on GPT-2 using 50 MRPC pairs (edge dim $k = 49$, using the $\min(64, d, n-1)$ cap in our script):

- Spectral H^0 dimension (eigenvalues $< 10^{-6}$): 49
- Algebraic H^0 dimension ($\dim C^0 - \operatorname{rank}(\delta^0)$ on the assembled coboundary matrix): 50
- Coboundary test: $\|\delta^0 \Pi_{H^0} s\| < 10^{-6}$ for all features tested

The only discrepancy is one eigenvalue near the cutoff ($\lambda \approx 4 \times 10^{-6} > 10^{-6}$).

B Additional Experiments

B.1 Cross-Dataset Generalization

We test cross-domain P for H^0 retrieval on the held-out CounterFact set (Table 9): we train the joint-PCA P on MRPC, PAWS or QQP paraphrases ($n = 300$) instead of CounterFact training facts, and apply it to the same CounterFact test set of centroids and queries. Across all 8 models in-domain P averages 81.6%, cross-domain P averages 55–60%, PCA-20 (no kernel) is 30.5% and Random-20 is 51.2%. Cross-domain P still benefits from kernel selection vs. no kernel ($+5$ to $+30$ pp) but lags behind in-domain P by $\sim$25 pp. There is some domain dependence in P. News (MRPC), adversarial (PAWS) and question (QQP) paraphrases all show non-trivial transfer, but a task that distinguishes facts benefits from P trained on facts. This is a limitation, and we recommend training P in-domain when the downstream retrieval domain is fixed.

Table 9: Cross-dataset restriction-map transfer (held-out CounterFact, hard same-relation retrieval, $k = 20$). H^0 is fit with P from the source on the left and evaluated on the same CounterFact test set. Out-of-domain P retains $+5$ to $+30$ pp over no-kernel baselines but loses $\sim$25 pp relative to in-domain P.

Model	H^0 **CF (in-domain)**	H^0 **MRPC**	H^0 **PAWS**	H^0 **QQP**	**PCA-20**	**Random-20**
GPT-2	85.3	55.1	63.9	59.1	24.6	39.1
Phi-2	87.1	57.3	61.6	57.6	37.2	55.1
Mistral-7B	87.1	57.8	65.9	61.2	34.5	59.2
OLMo-2-7B	82.8	64.1	60.5	57.6	35.7	62.3
Qwen-2.5-7B	64.1	47.0	53.3	56.7	37.5	43.7
Gemma-2-9B	77.9	48.3	47.9	51.0	27.5	43.5
Llama-2-7B	80.4	60.5	60.0	53.5	22.3	50.7
Llama-3-8B	88.0	56.9	65.5	60.5	24.4	55.9
Mean	**81.6**	55.9	59.8	57.1	30.5	51.2

B.2 Steering Magnitude Sweep

We vary $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$ on Mistral-7B and Llama-3-8B. With α in the middle range (0.10–0.15) the sheaf-derived methods outperform random steering by 1.0–2.0pp on both models. When we increase the magnitude ($\alpha \geq 0.30$), the concentrated H^1 influence over-perturbs the robust models and random becomes comparable. This leads to our recommendations in Appendix C.3.

B.3 Cycle-Aware H^0: Multi-Paraphrase Rings

In the body retrieval we use the pair-only graph: each fact has its paraphrases chained by $K-1$ edges, giving a forest with $b_1 = 0$. We want to see if connecting all paraphrases of a fact improves H^0 retrieval, and if closing a cycle helps as well.

Construction. For each fact with $K \in [4, 7]$ paraphrases (we cap at 5) we create a *multi-paraphrase ring*, a closed cycle through all K paraphrases of the fact. There are $N_{\text{train}} = 350$ facts and thus $\sum_i K_i$ edges and $b_1 = N_{\text{train}}$ cycles ($\sim$340 cycles). The kernel of $L_{\mathcal{F}}$ now requires invariance across all paraphrases of a fact, which is more restrictive than requiring invariance between pairs. We use the same node activations and joint-PCA P as before. Only the edges differ, and therefore the cycles.

Ring and clique both outperform pair-only on all architectures, by an average of $+6.5$ pp. Ring and clique perform within 0.1 pp of each other on average, so adding more edges than the ring does not help.

Edge-count-matched control. We want to disentangle the effect of jointly constraining all paraphrases of a fact from closing the cycle. On all facts with at least 6 paraphrases, we compare a ring over 5 nodes (5 edges, $b_1 = 1$) with an acyclic path over six paraphrases of the same fact (5 edges, $b_1 = 0$). Paired bootstrap confidence intervals on held-out facts show that the path beats pairs by

Table 10: Cycle-aware H^0 for hard-CounterFact retrieval. Both multi-paraphrase rings ($b_1 = N$) and full paraphrase cliques ($b_1 = N(K-1)(K-2)/2$) beat the pair-only acyclic forest ($b_1 = 0$) on all eight architectures, mean $+6.5$ pp. Ring and clique perform within 0.1 pp of each other.

Model	H^0 **pair** (b_1=0)	H^0 **ring** (b_1=N)	H^0 **clique** ($b_1 \gg N$)	Δ_{ring}	Δ_{clique}
GPT-2	87.4	90.7	89.2	$+3.4$	$+1.8$
Phi-2	83.1	89.6	90.5	$+6.5$	$+7.4$
Mistral-7B	86.7	89.8	91.4	$+3.2$	$+4.7$
OLMo-2-7B	84.4	91.4	91.4	$+7.0$	$+7.0$
Qwen-2.5-7B	62.3	71.8	70.7	$+9.5$	$+8.4$
Gemma-2-9B	75.2	84.0	84.7	$+8.8$	$+9.5$
Llama-2-7B	80.8	88.9	89.6	$+8.1$	$+8.8$
Llama-3-8B	86.5	92.3	90.7	$+5.9$	$+4.3$
Mean	**80.8**	**87.3**	**87.3**	$\mathbf{+6.5}$	$\mathbf{+6.5}$

$+7.7$ pp on average across the 8 models. The ring beats the path by $+0.5$ pp on average, and the 95% CI contains 0 for every one of the 8 models. So the gain comes from jointly constraining all paraphrases of a fact. The difference which closes the cycle is just the sum of the differences along the path, and it puts no extra constraint on exact global sections.

B.4 Restriction Map Sensitivity

We look at three different restriction maps: PCA (no parameters), CCA (maximize correlation) and contrastive (Fisher discriminant). We find that the best restriction map varies by architecture. CCA improves Mistral-7B by +2.2pp (n=1000, α=0.3), while contrastive achieves 31.0% vs. 4.2% for random on Llama-2-7B (7.4× increase, $p < 10^{-70}$). Figure 4 shows results for 3 of the 4 architectures we evaluated on CounterFact (Llama-2-13B's smaller method grid appears only in Table 4). The figure compares Random, CAA, PIXEL, sheaf H^1 and the best sheaf method for that architecture. The architecture-dependent pattern is visible across rows: sheaf wins outright on fragile models and is statistically tied with random on robust models.

Restriction-map ablation: PCA vs random vs identity. Table 11 contains the ablation study on restriction maps discussed in §5.2. We calculate H^0 using the same activations but with 3 methods: PCA-P (the current method, taking the top 128 PCs from joint-PCA), a random orthonormal P (5 seeds) and identity P (no projection; we use the bottom 20 eigenvectors of C_W in the full d-dimensional residual stream). We use hard-CounterFact retrieval and report top-1 accuracy with same-relation distractors, on $n_{\text{test}} = 150$ facts.

Table 11: Restriction-map ablation (8 architectures, hard-CounterFact retrieval). H^0 from $\ker L_{\mathcal{F}}$ recovers content directions whether P is joint-PCA, a random orthonormal projection, or absent.

Model	H^0 **PCA-**P	H^0 **random-**P	H^0 **identity-**P	**PCA-20** (no kernel)	**Random-20**
GPT-2	85.3	77.1 ± 0.8	79.0	24.6	39.1
Phi-2	87.1	73.0 ± 2.0	78.8	37.2	55.1
Mistral-7B	87.1	80.6 ± 1.6	83.3	34.5	59.2
OLMo-2-7B	82.8	76.4 ± 1.6	79.7	35.7	62.3
Qwen-2.5-7B	64.1	61.4 ± 1.3	62.3	37.5	43.7
Gemma-2-9B	77.9	69.2 ± 1.5	76.5	27.5	43.5
Llama-2-7B	80.4	76.3 ± 2.7	80.1	22.3	50.7
Llama-3-8B	88.0	80.0 ± 1.4	79.9	24.4	55.9
Mean	**81.6**	**74.2**	**77.5**	30.5	51.2

Identity-P (no data-dependent projection) lands within 4.1 pp of PCA-P on average, and beats PCA-without-kernel by $+47$ pp. Random-P averaged over 5 seeds lands within 8 pp of PCA-P. The kernel selection recovers the content, and the joint-PCA P shrinks the eigenproblem from $\mathbb{R}^d$ to $\mathbb{R}^{128}$. On its own, P does not select content directions.

B.5 Retrieval Baseline Comparison

We evaluate sheaf H^0 alongside vanilla subspace and representation-learning baselines on the same task (Table 12): LDA, CCA, PLS, top-20 PCA and metric learning (NCA). Following Table 2, we use the held-out hard-CounterFact setting with the same activations, 70/30 fact split, 20-dim budget and top-1 scoring. All methods except whitened PCA come from the same shared run. The sheaf H^0 and Full columns differ by under 1 pp from Table 2 (different seed and sampling). Whitened PCA (W-PCA) centers the activations then scales each top-20 principal direction by $1/\sqrt{\lambda}$. Its column is reported from an independent run following the same protocol. Whitening improves PCA-20 by at most 17 pp. Despite this improvement, W-PCA underperforms sheaf H^0 by 27 to 48 pp across architectures. LDA and NCA utilize the fact identity labels. Sheaf H^0, CCA, PLS, PCA-20 and W-PCA do not require labels. Sheaf H^0 outperforms every label-free method by $+9$ to $+36$ pp across all 8 models. Across all models, sheaf H^0 falls within $\sim$2 pp of the best label-requiring method on average (max difference is 8 pp, on Qwen) and outright wins on Phi-2, Gemma-2 and Llama-3. For reference only, we also include supervised sentence encoders trained on large amounts of paraphrase data (SimCSE 97.5%, a sentence transformer 99.1%). The sheaf operator also provides the H^1 influence subspace, the steering decomposition and the harmonic-mass fragility diagnostic computed from the same construction. LDA and NCA do not provide any of these signals.

Table 12: Retrieval baseline comparison (held-out hard-CounterFact, top-1 %, $k = 20$, same protocol as Table 2). Label-free: sheaf H^0, CCA, PLS, PCA-20 (top-20 principal components without kernel selection, as in Table 2), W-PCA (the same 20 components whitened, each scaled by $1/\sqrt{\lambda}$ on centred activations). Label-requiring: LDA, NCA. Sheaf H^0 bold where it beats both LDA and NCA.

Model	**Sheaf H^0**	**LDA**	**NCA**	**CCA**	**PLS**	**PCA-20**	**W-PCA**	**Full**
GPT-2	84.9	87.4	88.3	47.9	49.0	24.4	39.5	43.1
Phi-2	**87.6**	84.9	83.3	58.9	64.1	37.9	42.7	69.1
Mistral-7B	86.9	90.1	88.3	60.7	63.7	34.3	45.1	77.4
OLMo-2-7B	82.6	88.5	86.9	62.8	67.5	36.6	37.2	81.3
Qwen-2.5-7B	64.8	72.5	72.9	50.8	56.0	36.6	38.1	50.3
Gemma-2-9B	**77.4**	76.1	76.3	51.2	50.1	27.3	37.2	54.4
Llama-2-7B	80.6	83.5	84.4	49.7	49.9	21.9	39.3	68.2
Llama-3-8B	**87.8**	86.5	86.0	53.3	55.5	24.4	39.3	72.7
Mean	81.6	83.7	83.3	54.4	57.0	30.4	39.8	64.6

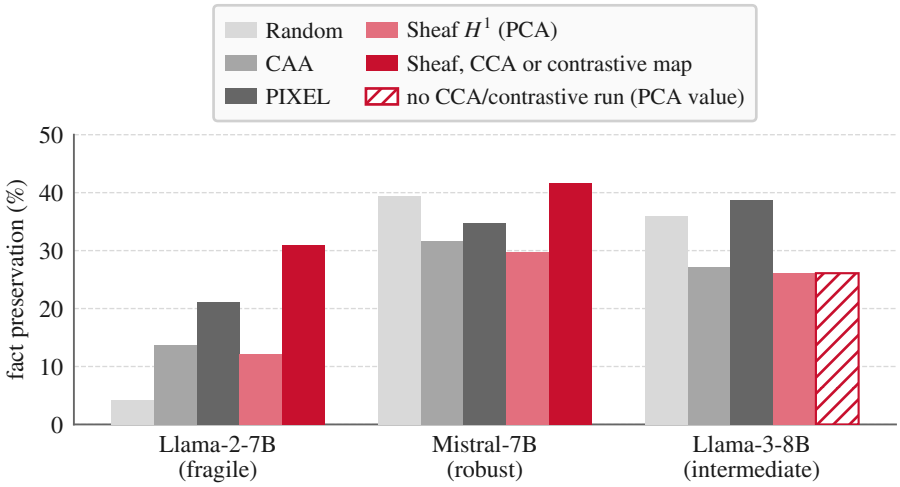


Figure 4: Steering on CounterFact at $\alpha = 0.3$ ($n = 1000$ per model) for the three architectures with a PIXEL run. The hatched bar on Llama-3-8B repeats the PCA H^1 value, since that model has no CCA or contrastive run. On fragile Llama-2-7B, random steering collapses to 4.2% and the contrastive sheaf leads at 31.0%. On robust Mistral-7B, the CCA sheaf and random tie (41.6% vs. 39.4%). On Llama-3-8B, PIXEL and random lead. Same data as Table 4 and Appendix B.11.

B.6 Sheaf Eigenbasis Ablation

This ablation uses the eigenbasis of $L_{\mathcal{F}}$ directly. At every other layer, H^1 is the top 20 eigenvectors of $L_{\mathcal{F}}$ with the joint-PCA restriction map ($k = 128$), fit on mean-pooled block outputs of 200 MRPC paraphrase pairs, and H^0 is the bottom 20. On 100 held-out MRPC sentences, the block output is projected onto the orthogonal complement of the subspace at every position, the rest of the forward pass runs, and the effect is the KL divergence of the next-token distribution at the final position. The control is 20 Haar-random 20-dimensional subspaces, which are not matched to H^1 in activation energy; Table 3 is the variability-matched comparison. Table 13 reports, per model, the ratio of mean KL (H^1 over random) as the median and range over layers, the smallest fraction of prompts over layers for which H^1 exceeds all 20 random draws, and the median H^0 ratio.

Table 13: Sheaf eigenbasis ablation: ratio of mean next-token KL from projecting out H^1 (top 20 eigenvectors of $L_{\mathcal{F}}$) to that from 20 random 20-dimensional subspaces at every other layer. Median and range are over layers. The prompts column is the smallest fraction over layers of held-out prompts on which H^1 exceeds all 20 draws.

Model	Layers	H^1/random, median	Range	Prompts	H^0/random, median
GPT-2 (124M)	6	17.8×	13.7–53.5	99%	0.19
Phi-2	16	84.1×	18.4–122	97%	0.56
Mistral-7B	16	398×	70.0–3075	100%	0.54
Qwen-2.5-7B	14	127×	78.7–293	99%	0.45
OLMo-2-7B	16	182×	112–1048	100%	2.21
Gemma-2-9B	21	240×	75.8–1211	100%	0.74
Llama-2-7B	16	287×	102–2283	100%	0.32
Llama-3-8B	16	176×	75.7–409	100%	0.29
Llama-2-13B	20	327×	191–645	100%	0.47

B.7 Dormancy and the Subspace-Patching Illusion

Makelov et al. [45] analyze an illusion for *interchange* patching, $a + (p_A - p_B)v$ with $v = (v_{\text{disc}} + v_{\text{dormant}})/\sqrt{2}$: activating variation along a causally disconnected direction can turn on a dormant direction. By contrast, Section 5.3 zero-ablates the subspace via projection onto its orthogonal complement. We conduct two checks to rule out a dormant-pathway interpretation of our results. By definition a dormant direction is one that sees no differential activation on-distribution. We compute the on-distribution projection spread of H^1, H^0, and five variance-matched random subspaces (same dimension, norm, and eigenspectrum). We then ablate each and report fact recall from the greedy 50-token continuation of the bare CounterFact prompt. This prompt does not contain the answer ($n = 100$ for GPT-2 and 150 for the 7B models). H^1 is the least dormant subspace we examine, and ablating it collapses fact recall, while ablating the matched random subspace leaves recall at baseline (Table 14). GPT-2 only recalls 15% of these facts unaided, so its row is the weakest. Because the generated text itself is modified by this intervention, the effect is behavioural and does not rely on the KL tail. We observe an intervention KL ratio of 15.7× on GPT-2. On the 7B models, the KL saturates numerically (median raw KL of 7.1 vs. 0.003 nats on Mistral-7B), so here we rely on the collapse of recall as our evidence.

Table 14: Dormancy and behavioural ablation. Dormancy ratio is the on-distribution projection spread of H^1 divided by the mean spread of variance-matched random subspaces. Recall is the fraction of greedy continuations of the bare prompt that contain the target entity. Layer $\ell \approx 2L/3$, $k = 20$, five random draws, seed 42.

Model	Dormancy ratio (H^1/random)	Recall, no ablation	H^1 ablated	Random ablated
GPT-2	2.5×	15.0%	2.0%	7.5%
Mistral-7B	6.6×	46.7%	0.0%	47.0%
Llama-2-7B	5.6×	47.3%	4.7%	48.7%

B.8 Cross-Lingual Invariance Probe

Nothing in the construction is paraphrase-specific: it needs a grouping of contexts asserted to be equivalent and their activations. As a transfer check we use English–French translation pairs from OPUS-100 as sheaf edges on Llama-3-8B (layer 21, edge dimension 128, $k = 20$, seed 42), fit H^0 on 100 pairs, and evaluate top-1 translation retrieval on 100 disjoint held-out pairs (Table 15). The H^0 kernel retains cross-lingual retrieval and matches or exceeds the full 4096-dimensional hidden state in the EN→FR direction, while the H^1 complement discards it. The decomposition separates translation-invariant content from language-specific variation as it does for paraphrase; a full cross-lingual study is future work.

Table 15: Held-out cross-lingual retrieval on Llama-3-8B (top-1 %, 100 test pairs, H^0 fit on 100 disjoint training pairs).

Subspace	FR→EN	EN→FR
H^0 kernel (20-dim)	76	78
H^1 complement	6	3
Full hidden state (4096-dim)	82	59

B.9 Per-Architecture Layer Profiles

For our three-depth runs (Table 3, protocol b) we look at the H^1-to-control ratio at $L/4$, $L/2$ and $3L/4$ for all nine architectures (Figure 5). The depth with the highest ratio depends on the architecture. GPT-2, Mistral-7B, Llama-3-8B, Llama-2-7B, Llama-2-13B, Phi-2 and Qwen-2.5-7B peak at $L/4$, while OLMo-2-7B and Gemma-2-9B peak at $L/2$. Three depths per model cannot resolve a layer-wise shape. For GPT-2, Llama-2-7B, Llama-3-8B and Llama-2-13B, the table reports our separate 1000-sentence run at $L/2$ (protocol a).

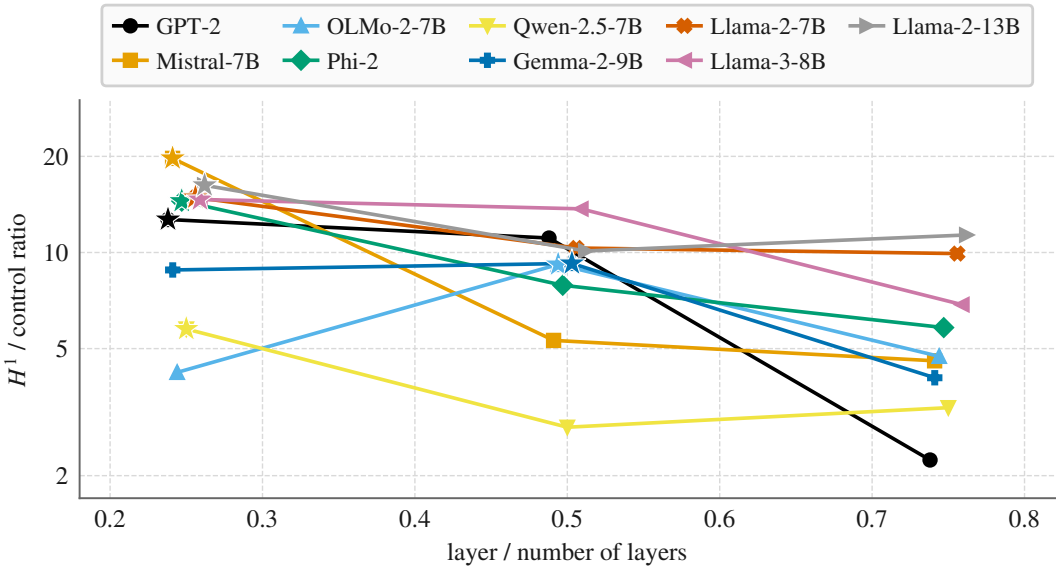


Figure 5: H^1-to-control effect ratio (protocol b of Table 3) at three depths per architecture, $L/4$, $L/2$ and $3L/4$, plotted against normalized depth (layer index over number of layers). Stars mark each architecture's largest measured ratio, and points are offset slightly in depth so overlapping peaks stay visible. Only measured points are shown. The GPT-2 and Llama values here come from protocol b and differ from their protocol-a rows in Table 3.

B.10 McNemar Tests with Multiple Comparisons Correction

All main comparisons (sheaf vs. random, sheaf vs. Fisher) are evaluated via McNemar's test (n=1000 per model) with Benjamini-Hochberg correction for 16 simultaneous tests ($q < 0.05$). 13

of 16 tests survive correction, with the largest effects on Llama-2-7B (Cohen's $h = 2\arcsin\sqrt{p_1} - 2\arcsin\sqrt{p_2} = +0.35$, $p < 0.001$).

B.11 PIXEL Head-to-Head on CounterFact

PIXEL [66] (Yu et al., 2025) extracts steering directions per layer using two complementary views of the prompt: tail-averaged hidden states (mean over the last $k_{\text{tail}} = \text{clip}(\text{round}(0.10 \cdot \text{plen}), 3, 8)$ prompt tokens) and the end-token hidden state. It then takes the differences between paired positive and negative samples, stacks them across the two views, and uses the top weighted PCA direction (r=1) as the unit vector u_ℓ for each layer. At inference PIXEL computes the magnitude for each (layer, prompt-position) site with the following closed form:

$$a = h^\top u, \quad B = \sqrt{\|h\|^2 - a^2}, \quad T = Bs/\sqrt{1 - s^2},$$
$$\alpha = \text{clip}\big(\max(\max(0, -a/c),\ (T - a)/c),\ [\alpha_{\min}, \alpha_{\max}]\big),$$

where $s = z_{\text{target}}$ is the target z-score (we use the PIXEL default $s = 0.85$) and c is the original direction norm (here $c = 1$).

We reimplement PIXEL's direction extraction and per-position magnitudes as published, but we use one global direction (extracted once from the casual/formal CAA pairs) instead of a direction for each question. CounterFact has no per-question gating signal similar to PIXEL's MC2 score on TruthfulQA, so we cannot extract per-question directions. We inject only at the last k_{tail} positions of the prompt during the prompt forward pass. The repetition penalty (1.3), greedy decoding, 100 max new tokens, layer $\ell = \lfloor 5L/8 \rfloor = 20$ and 4-bit quantization all match §5.4. The only difference between PIXEL and our other methods is how directions are extracted and how per-position magnitudes are calibrated.

We run PIXEL at $n = 1000$ with $\alpha_{\text{clip}} = [0, 200]$, which gives the closed form full freedom (results in Table 4). Similar to our methods, PIXEL strongly improves over CAA on the fragile Llama-2-7B and is approached or beaten by random on the robust Mistral-7B. This further supports the architecture-diagnostic reading: the influence-concentration mechanism (§5.4) is shared across direction-extraction recipes. Using two views to extract directions is better than taking the mean of differences like CAA does, but calibrating per-position magnitudes in closed form still does not overcome architecture fragility. The strongest single method on Llama-2-7B is still contrastive sheaf steering with the architecture-matched restriction map (31.0%). On Mistral-7B, CCA sheaf steering is strongest (41.6%). No sheaf method was able to beat random on Llama-3-8B at $\alpha = 0.30$.

What we didn't do: PIXEL applies a per-question soft gate using a labeled MC2 score on TruthfulQA, which is not available for CounterFact, so we apply one global direction. PIXEL has an option to inject at continuation positions ("include-pred"), which would require us to modify activations during incremental decoding with the KV cache. This would be a bigger change than the direction-vs-direction comparison we are trying to make here. PIXEL also tunes residual_scale and auto-flip for each benchmark. We leave these at their default values (1.0 and no flip) and use sign-aligned PCA. These changes all deal with the magnitude side of PIXEL. We follow their direction-extraction recipe, which is the part we want to compare to our H^1 direction.

B.12 Holonomy Null Calibration

The holonomy around a cycle γ is the composition H_γ of the orthogonal transports along that cycle's edges. The obstruction satisfies $\|I - H_\gamma\|_F^2 = 2k - 2\,\text{tr}\,H_\gamma \leq 4k$, so it's bounded by $2\sqrt{k}$, and a Haar-random orthogonal matrix gives $\mathbb{E}\|I - H_\gamma\|_F^2 = 2k$ (RMS of $\sqrt{2k}$), so raw holonomy magnitudes need calibrating with the nulls below.

We calibrate the cycle-holonomy obstruction $\|I - H_\gamma\|_F$ against two nulls and a finite-sample stability check (full data in Table 16). All numbers use edge dim $k = 8$, 11 fundamental cycles, layer $\ell = 8$ for GPT-2 and $\ell = 16$ for the others, and 6 token samples per node.

Shuffled-paraphrase null. For each of 20 seeds, randomly permute which paraphrase belongs to which fact *within each relation*. Refit per-node PCA P_v and Procrustes transports Q_{uv} on the shuffled assignments. Recompute holonomy. Observed obstructions sit 0.10–0.29 above the shuffled null on all four architectures, and none of the differences is significant (one-sided $p = 0.10$–0.33).

Bootstrap on tokens-per-node. Refit P_v and Q_{uv} using $N \in \{2, 3, 4, 5, 6\}$ token samples per node. On all four architectures the obstruction falls as N grows (from 3.1–4.4 at $N = 2$ to 2.2–2.8 at

$N = 6$), so part of the observed magnitude is finite-sample noise in the Procrustes fits, which are underdetermined when $N < k$.

Identity-transport sanity. Setting every $Q_{uv} = I$ gives $\|I - H_\gamma\|_F = 0$ to machine precision, and on synthetic flat data, where every true holonomy is I, the pipeline recovers $\|I - H_\gamma\|_F < 10^{-4}$.

Table 16: Holonomy null calibration on the relation hypergraph. "Observed" = real paraphrase assignments. "Shuffled null" = mean over 20 within-relation paraphrase shuffles. p is the one-sided permutation p-value for $P(\text{shuffled} \geq \text{observed})$, smaller meaning observed obstruction exceeds null. Bootstrap-N obstructions are mean $\pm$ std over 11 cycles at the listed token count.

	Observed	**Shuffled**	p	N=2	N=3	N=4	N=6
GPT-2 (124M)	2.92	2.78	0.24	3.09	3.15	2.55	2.43
Mistral-7B	2.52	2.42	0.33	3.20	3.31	2.64	2.18
Qwen-2.5-7B	3.13	2.84	0.10	3.82	3.43	2.90	2.78
Phi-2	3.01	2.76	0.24	4.37	3.37	3.28	2.36

Cycle holonomy on the relation hypergraph is therefore far from Haar-random but not distinguishable from a shuffled-paraphrase null at this sample size: the transports carry relation structure that survives reassigning paraphrases within a relation, and no architecture shows a fact-specific holonomy signal.

B.13 Node-Space Operationalizations of Relation-Hypergraph H^1

As shown in §4.2, our relation hypergraph has $\dim H^1 = 66$ (algebraically) and 18.4% Hodge harmonic mass. Lifting this to node space did not give a downstream benefit.

Harmonics as features. We tried projecting our node activations onto the harmonic subspace $\ker(L_1)$, pulled back to the node stalks via the δ^0 adjoints. This gave us a retrieval rate of 36.3% on our 60-fact problem (§5.2, with hard negatives from the same relation). This was lower than our random-20 baseline, which scored 49.1% on GPT-2. Harmonics represent where something cannot be glued together, not facts. Directions that are stable around cycles are stable across a relation but not within a fact, so the basis does not match the retrieval target.

We can still say that the algebraic result holds. As seen in Appendix A.1, the spectrum and Hodge mass establish it on their own. Projecting relation-hypergraph H^1 to node space did not give a retrieval benefit. Untested options include cycle-aware steering, supervised features learned on top of the harmonic basis, and per-node restriction maps trained for downstream prediction.

B.14 Sheaf-Regularized SSL Collapse Prevention (Per-Architecture)

These are the results for individual architectures referenced in §7. Using an MLP with 2 hidden layers ($d \to 256 \to 256 \to 64$) we train on 300 MRPC paraphrase pairs for 1500 Adam steps. With either naive MSE or $\mathrm{tr}(L_{\mathcal{F}})$ alone, the encoder degenerates to an effective rank of 14–42 out of 64 dimensions, with pair MSE indistinguishable from zero. When using $\mathrm{tr}(L_{\mathcal{F}}) - \beta \log\det \mathrm{Cov}$ we see effective ranks between 57–60 out of 64 for all seven architectures.

B.15 MLST on STS-B (Full Table)

Multi-Layer Sheaf Consistency Training compared against frozen baseline, SimCSE, and VICReg on STS-B Spearman/Pearson. Frozen encoders, edge dim $k = 64$; trainable per-layer projections P_l and transitions $T_{l \to l+1}$; 2000 Adam steps on MRPC paraphrase pairs (positive labels only). GPT-2 with layers $\{3, 6, 9, 12\}$ and 10 seeds; Mistral-7B with layers $\{8, 16, 24, 32\}$ and 5 seeds; same hyperparameters in both. The *nosheaf* ablation drops the T-transitions and the cycle-holonomy term, keeping only per-layer paraphrase consistency with the variance/covariance regulariser. Read-out modes: *final* = $P_L^\top h^L$; *concat* = $[P_l^\top h^l]_l$ stacked; *section* = concat together with T-transported predictions $T_{l-1 \to l} P_{l-1}^\top h^{l-1}$.

Paired-t tests on per-seed Spearman, GPT-2 ($n = 10$): MLST-nosheaf-concat $-$ VICReg $\Delta = +0.046$, $p = 6 \times 10^{-9}$; MLST-section $-$ VICReg $\Delta = +0.035$, $p = 1 \times 10^{-6}$; MLST-section $-$ SimCSE $\Delta = +0.080$, $p = 2 \times 10^{-7}$; MLST-section $-$ MLST-nosheaf-concat $\Delta = -0.010$,

Embedding	Loss	Top-1 σ^2 frac.	Effective rank	Feature var.	Pair MSE
GPT-2 (768D)	Naive MSE	0.149	16.0/64	0.006	0.0000
	$\mathrm{tr}(L_{\mathcal{F}})$ only	0.151	16.6/64	0.008	0.0000
	$\mathrm{tr}(L_{\mathcal{F}}) - \beta \log\det \mathrm{Cov}$	0.109	**58.5/64**	**4991**	0.0149
Mistral-7B (4096D)	Naive MSE	0.134	34.4/64	0.005	0.0000
	$\mathrm{tr}(L_{\mathcal{F}})$ only	0.121	33.1/64	0.006	0.0000
	$\mathrm{tr}(L_{\mathcal{F}}) - \beta \log\det \mathrm{Cov}$	0.094	**58.0/64**	**8418**	0.0186
Qwen-2.5-7B (3584D)	Naive MSE	0.107	35.9/64	0.006	0.0000
	$\mathrm{tr}(L_{\mathcal{F}})$ only	0.119	27.5/64	0.005	0.0000
	$\mathrm{tr}(L_{\mathcal{F}}) - \beta \log\det \mathrm{Cov}$	0.143	**58.5/64**	**6213**	0.0202
Llama-2-7B (4096D)	Naive MSE	0.155	36.1/64	0.006	0.0000
	$\mathrm{tr}(L_{\mathcal{F}})$ only	0.115	42.5/64	0.006	0.0000
	$\mathrm{tr}(L_{\mathcal{F}}) - \beta \log\det \mathrm{Cov}$	0.082	**59.5/64**	**6512**	0.0128
Llama-3-8B (4096D)	Naive MSE	0.169	22.2/64	0.006	0.0000
	$\mathrm{tr}(L_{\mathcal{F}})$ only	0.115	33.6/64	0.006	0.0000
	$\mathrm{tr}(L_{\mathcal{F}}) - \beta \log\det \mathrm{Cov}$	0.109	**58.0/64**	**6843**	0.0316
OLMo-2-7B (4096D)	Naive MSE	0.168	23.3/64	0.006	0.0000
	$\mathrm{tr}(L_{\mathcal{F}})$ only	0.420	14.3/64	0.005	0.0000
	$\mathrm{tr}(L_{\mathcal{F}}) - \beta \log\det \mathrm{Cov}$	0.184	**57.0/64**	**4127**	0.0134
Gemma-2-9B (3584D)	Naive MSE	0.185	23.4/64	0.006	0.0000
	$\mathrm{tr}(L_{\mathcal{F}})$ only	0.145	23.0/64	0.006	0.0000
	$\mathrm{tr}(L_{\mathcal{F}}) - \beta \log\det \mathrm{Cov}$	0.104	**57.8/64**	**3686**	0.0133

Table 17: Collapse prevention per architecture: top-1 variance fraction, effective rank (of 64), feature variance and paraphrase-pair MSE under naive MSE, the Laplacian energy $\mathrm{tr}(L_{\mathcal{F}})$ alone, and the Laplacian energy with a log-determinant covariance term.

Method	**GPT-2** ($n = 10$) Spearman ↑	Pearson ↑	**Mistral-7B** ($n = 5$) Spearman ↑	Pearson ↑
Vanilla (no training)	0.428	0.312	0.533	0.498
SimCSE (final-layer, InfoNCE)	0.578 ± 0.014	0.567	0.566 ± 0.016	0.549
VICReg (final-layer, inv + var + cov)	0.623 ± 0.006	0.602	$\mathbf{0.645 \pm 0.025}$	**0.635**
MLST nosheaf (final)	0.621 ± 0.005	0.598	0.634 ± 0.025	0.626
MLST nosheaf (concat)	$\mathbf{0.668 \pm 0.005}$	**0.650**	0.644 ± 0.009	0.630
MLST full (final)	0.620 ± 0.006	0.597	0.622 ± 0.062	0.616
MLST full (concat)	0.658 ± 0.007	0.635	0.631 ± 0.033	0.618
MLST full (section)	0.658 ± 0.006	0.634	0.629 ± 0.021	0.611

Table 18: Multi-layer sheaf consistency training (MLST) on STS-B with frozen encoders: Spearman and Pearson correlation, mean $\pm$ s.d. over seeds. *nosheaf* drops the transitions and the cycle term.

$p = 9 \times 10^{-4}$. Mistral-7B ($n = 5$): MLST-section − VICReg $\Delta = -0.016$, $p = 0.20$; MLST-nosheaf-concat − VICReg $\Delta = -0.001$, $p = 0.94$; MLST-section − SimCSE $\Delta = +0.063$, $p = 0.005$. Cross-architecture rank-correlation across the eight trained methods is $\rho = 0.74$, $p = 0.037$. The robust positive finding on both architectures is multi-layer projection + variance/covariance regularization > single-layer baselines; the specific sheaf transition and cycle apparatus is neutral-to-slightly-negative as a training signal at this scale. The cycle-holonomy term is too easily satisfied by paraphrase-mean alignment and overlaps with the consistency loss; the Q-transitions add no consistent gradient signal beyond what concat already provides. Larger encoders, harder tasks, or richer cycle structure could change this; we leave that to future work.

C Reproducibility

C.1 Hyperparameter Sensitivity

The spectral gap (ratio of maximum to minimum eigenvalues) stays in 91.7–95.6× across edge dimensions $k \in \{32, 64, 128\}$ (the count m selects eigenvectors and does not change the spectrum); the content/context boundary is stable under these choices.

m sensitivity for H^0 retrieval. We sweep $m \in \{5, 10, 20, 40, 80, 128\}$ (with edge dimension $k = 128$ fixed) on hard CounterFact retrieval (Table 19). The plateau spans $m \in [20, 80]$ on all four tested architectures; $m = 40$ is consistently slightly better than $m = 20$ ($+4$ to $+6$ pp). The boundary case $m = 128 = k$ leaves no eigenvectors above the spectral gap (no kernel selection) and collapses to 36–53%. The full hidden state without projection lands at 43–78%. We chose $m = 20$ *a priori* to leave a comfortable margin from the k boundary; results are not test-tuned and are stable across the plateau.

Table 19: H^0 subspace dimension m sweep (hard CounterFact retrieval, edge dim $k = 128$). Results are stable across $m \in \{20, 40, 80\}$. At $m = 128 = k$ no kernel is selected and retrieval collapses.

Model	m=5	m=10	m=20	m=40	m=80	m=128	**Full**
GPT-2	63.7	74.3	85.3	**91.2**	87.8	35.9	43.1
Phi-2	67.3	75.8	87.1	**90.3**	85.3	52.8	69.1
Mistral-7B	70.9	82.4	87.1	**90.3**	86.7	48.5	77.7
Llama-2-7B	62.8	74.0	80.4	**85.3**	85.1	35.2	68.2

Pooling sensitivity. We compare three pooling choices (mean over all tokens, last non-pad token, first token) by extracting activations under each and rerunning the full H^0 pipeline (Table 20). Mean pooling is best across all four architectures (H^0 at 80–87%); last-token pooling drops 10–30 pp; first-token pooling collapses (autoregressive models have very limited content at position 0). Sheaf H^0 beats the matched Full hidden state under mean and last-token pooling on every architecture, and under first-token pooling on three of four (on Mistral-7B, 14.2% against 20.5%). Absolute retrieval rates depend on whether the chosen pooling captures content.

Table 20: Pooling sensitivity (hard CounterFact retrieval). Mean pooling is best for all architectures, and sheaf H^0 beats the full hidden state under every pooling choice except first-token pooling on Mistral-7B.

	Mean (default)		**Last token**		**First token**	
Model	Full	H^0	Full	H^0	Full	H^0
GPT-2	43.1	**85.3**	25.5	74.3	41.8	42.7
Phi-2	69.1	**87.1**	26.9	65.9	30.5	43.8
Mistral-7B	77.7	**87.1**	40.4	72.2	20.5	14.2
Llama-2-7B	68.2	**80.4**	21.4	47.0	19.6	23.0

C.2 Experimental Details

Compute: All experiments on NVIDIA A100 or RTX 4090 GPUs with 80GB or 24GB VRAM respectively. 7B+ models use 4-bit quantization (bitsandbytes). Sheaf construction (restriction map + Laplacian) takes < 1 minute per model; dominant cost is activation extraction (identical to any activation-based method).

Hyperparameters: Edge dimension $k = 128$ (capturing $\geq 98\%$ of variance). Mean pooling over sequence positions. Restriction map learned via joint PCA. Top/bottom 20 eigenvectors define H^0/H^1. Steering vectors norm-matched to 30% of mean hidden state norm.

Layer selection: retrieval and the sheaf constructions use $\ell = \lfloor 0.67L \rfloor$ (layer 8 for GPT-2, layer 21 for 32-layer models), and steering uses $\ell = \lfloor 5L/8 \rfloor$ (layer 7 for GPT-2, layer 20 for 32-layer models). Targets middle-to-late layers where semantic content is well-formed but output projections have not dominated.

Statistical tests: Bootstrap confidence intervals (1000 resamples). Mann-Whitney U for group comparisons. McNemar for paired binary outcomes. Cohen's d (continuous) and Cohen's h (proportions) for effect sizes. All p-values are two-sided except the Spearman and holonomy permutation tests, which are exact one-sided in the pre-specified direction.

C.3 Practitioner Guidance

Steps to follow when using this method. (1) Create your sheaf using at least 200 paraphrase pairs from MRPC. Use PCA to create the restriction maps (no parameters to tune by default). (2) Then steer at an α around 0.10–0.15 using the H^1 dims and H^0-removed. Choose whichever performs best on a validation set of size 50. (3) If one random-steering probe shows your model is susceptible to generation collapse, then use the sheaf H^1 to prevent this. If your model is robust, random steering can perform similarly to sheaf steering at $\alpha \geq 0.30$, so stay at moderate values to keep the sheaf advantage.

C.4 NeurIPS Paper Checklist

1. **Claims**
   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   Answer: [Yes]
   Justification: Section 1 enumerates three contributions, with the SSL result reported in §7: (i) H^0 beats LEACE on held-out CounterFact, the kernel selection carries the result under random or identity P, and joint multi-paraphrase constraints add $+6.5$ pp (Tables 2, 11, 10); (ii) H^1 functional influence (5.6–26.5$\times$ over coordinates of matched per-coordinate variability; Table 3); (iii) multi-layer grids carry a harmonic mass that ranks six architectures by steering fragility at $\rho = 0.90$ (Table 1, Appendix A.3). Each claim is supported by experiments in §5 with effect sizes, p-values, and bootstrap CIs. The negative result on MLST cycle/transition terms is reported in Appendix B.15 (Table 18).

2. **Limitations**
   Question: Does the paper discuss the limitations of the work performed by the authors?
   Answer: [Yes]
   Justification: Section 7 contains an explicit Limitations paragraph covering: (a) restriction-map selection is architecture-dependent (contrastive Fisher wins on Llama-2-7B, CCA on Mistral-7B); (b) restriction-map selection is partially domain-dependent (cross-corpus P costs $\sim$25 pp, Appendix B.1); (c) sheaf steering wins at $\alpha \in [0.10, 0.15]$ but ties random at $\alpha = 0.30$ on robust models; (d) a node-space operationalization of relation-hypergraph H^1 did not transfer to retrieval (Appendix B.13); (e) cycle holonomy does not differ significantly from a shuffled-paraphrase null (Appendix B.12); (f) the MLST cycle/transition terms are neutral-to-slightly-negative as a training signal (Appendix B.15); (g) our PIXEL comparison uses a simplified port (single global direction; Appendix B.11); we do not empirically compare to Angular Steering or MSRS (orthogonal frameworks).

3. **Theory, Assumptions and Proofs**
   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
   Answer: [Yes]
   Justification: The paper has one Proposition (Proposition 3.1: reduction of the sheaf Laplacian to projected within-pair covariance). The assumptions are stated explicitly in the proposition body (shared-projection matching-graph regime, one P for every node, $Q_{uv} = I$ on edges); the failure regime under per-node P_v and Procrustes transports is also stated. The full proof is provided in Appendix A.5. Remark 3.3 clarifies the algebraic-vs-spectral H^1 distinction so the cohomological language is unambiguous.

4. **Experimental Result Reproducibility**
   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?
   Answer: [Yes]
   Justification: The pipeline is specified in equation form in §3 (joint-PCA restriction map,

sheaf Laplacian as projected within-pair covariance, H^0/H^1 as smallest/largest eigenspaces, mean-difference steering with H^0-removal). All hyperparameters (edge dimension $k = 128$, H^0/H^1 count $m = 20$, layer $\ell \approx 2L/3$ ($\lfloor 5L/8 \rfloor$ for steering), steering norm $\alpha = 0.3\|\bar{h}\|$, MLP encoder geometry $d \to 256 \to 256 \to 64$, training step counts) are documented in §5 and Appendix C.2. Datasets are public (MRPC, PAWS, QQP, STS-B, CounterFact) with standard splits and 70/30 train/test for the LEACE comparison. Quantization (4-bit bitsandbytes for 7B+ models) and layer selection rules are explicit.

5. **Open Access to Data and Code**
   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?
   Answer: [Yes]
   Justification: All datasets used are public benchmarks with standard licenses (cited in §5 and below). Code implementing the sheaf decomposition, steering pipeline, multi-layer grid, and MLST, with a claim-by-claim reproduction guide, is released at `https://github.com/CWRU-AISM/gluing-lc`. The construction in §3 together with the hyperparameters in Appendix C.2 is sufficient to reimplement the pipeline from the paper alone, and all numerical results are deterministic given the published hyperparameters and seeds.

6. **Experimental Setting/Details**
   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?
   Answer: [Yes]
   Justification: Section 5 (Setup and Datasets) and Appendix C.2 specify: nine architectures with sizes; layer selection rule $\ell \approx 2L/3$ ($\lfloor 5L/8 \rfloor$ for steering); edge dimension $k = 128$ chosen to capture $\geq 98\%$ paired variance; H^0/H^1 slice $m = 20$; bootstrap CIs with 1000 resamples; McNemar's test for paired comparisons; restriction maps (PCA / CCA / contrastive Fisher) and how each is fit; 200 MRPC pairs for sheaf construction; CounterFact $n = 1000$ for steering; 70/30 fact split for LEACE comparison.

7. **Experiment Statistical Significance**
   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?
   Answer: [Yes]
   Justification: Bootstrap 95% CIs are reported alongside every steering and retrieval point estimate (Tables 3, 4, 2). McNemar's exact test is used for paired binary outcomes; Cohen's d for effect sizes; Mann–Whitney U for unpaired comparisons. Multiple-comparisons control via Benjamini–Hochberg is detailed in Appendix B.10. MLST results are over ten seeds with reported standard deviations and paired t-tests. All p-values are two-sided except the Spearman and holonomy permutation tests, which are exact one-sided in the pre-specified direction.

8. **Experiments Compute Resources**
   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?
   Answer: [Yes]
   Justification: Appendix C.2 reports hardware (NVIDIA A100 80GB or RTX 4090 24GB), 4-bit quantization for 7B+ models, and execution time (sheaf construction < 1 minute per model; activation extraction is the dominant cost). Section 5 indicates the cluster setup; no experiment requires more than a single A100.

9. **Code of Ethics**
   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?
   Answer: [Yes]
   Justification: The work is interpretability research on publicly released language models using public paraphrase and factual-recall benchmarks. No human subjects, no crowdsourcing, no scraping of private data, no release of new high-risk artifacts. Datasets used (MRPC, PAWS, QQP, STS-B, CounterFact) are standard licensed benchmarks. We have read the NeurIPS Code of Ethics and conform.

10. **Broader Impacts**
Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?
Answer: [Yes]
Justification: The work is interpretability research aimed at improving transparency of large language model representations. Positive impact: a label-free decomposition that exposes which directions in residual streams are content-stable across paraphrasing and which carry surface variation, potentially aiding red-teaming, alignment audits, and the diagnosis of model fragility (the steering analysis in §5.4 is itself a fragility diagnostic). Negative impact: the same machinery could in principle be used to construct targeted style or stance interventions against safety guardrails. We frame steering as a diagnostic tool rather than a deployment mechanism, and we report architecture-dependent failure modes (random steering can outperform sheaf steering at $\alpha = 0.30$ on robust models, which curbs naive misuse).

11. **Safeguards**
Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?
Answer: [NA]
Justification: The paper does not release new pretrained models, generative models, or scraped datasets. We analyze existing publicly released language models (GPT-2, Llama-2/3, Mistral, Qwen, phi-2, Gemma, OLMo) using existing public benchmarks. The released artifact is analysis code, not a model checkpoint or dataset.

12. **Licenses for existing assets**
Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?
Answer: [Yes]
Justification: All datasets and models are cited at first use: MRPC [15], PAWS [68], QQP [36], CounterFact [47], TruthfulQA [44]. Models are accessed through their official Hugging Face releases under their respective licenses (Apache-2.0 for Mistral and Qwen; the Llama-2/3 community license; MIT for phi-2; Apache-2.0 for OLMo; Gemma terms of use; MIT for GPT-2). Each is used within the terms permitted for academic research.

13. **New Assets**
Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?
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Justification: The paper introduces no new datasets, no new pretrained models, and no new benchmarks. All releasable artifacts are analysis code; the code repository documents them with a claim-by-claim reproduction guide.

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Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?
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Question: Does the paper describe the usage of LLMs if it is an important, original, or

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